



Northumbria University NEWCASTLE

KV6013 Computing Project Dissertation

BSc (Hons) Computer Science

May 2026

Evaluating the Effectiveness of a Voice-Interactive AI-Based Learning System Compared to Traditional E-Textbook Methodologies in Astronomy

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Practical Work One Drive Link: https://livenorthumbriaac-my.sharepoint.com/:f/g/personal/w23012928_northumbria_ac_uk/IgAfvs1h68M0QaW01MWNAWEuAahdszqqSOY9iq-6k0tPdGA?e=jtQCDC

Word Count: 13,200

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Zoe Moorton. Without her guidance and support throughout the development of my project, this work would not have been possible. I am grateful for the continuous dedication and encouragement she has provided, as well as her consistent responsiveness.

To my close family, Mum, Dad and Fran, I am deeply grateful for your constant belief in me and my abilities. Thank you for your consistent support throughout the project, especially through our phone calls and messages.

To my close friends throughout university, thank you for your understanding and support. Without you, I would not be here studying a course I truly enjoy. From the late nights across Europe to Newcastle, thank you for supporting me and always being there.

To my dogs, Romeo and Brodie, thank you for your comfort throughout the project. The much-needed breaks were far better with you by my side.

And finally, to Nadiya, my greatest supporter, my rock and my source of strength. Without your continuous support, this project would not have been possible. Thank you for standing by me and for your constant encouragement; I am sincerely grateful.

Abstract

Astronomy education faces several core challenges, including learners struggling to conceptualise astronomical phenomena. This research proposed a potential solution by developing a voice-interactive artificial intelligence (AI) astronomy application, which was then compared to traditional instructional approaches, specifically e-textbooks. A Randomised Controlled Trial (RCT) crossover method was conducted on a small sample of university students (n=8) to evaluate the learning outcomes between the pedagogical approaches. The findings demonstrated that the developed AI system outperformed the traditional instructional approach within this study's constrained timeframe. However, the collected qualitative participant feedback revealed that there was a discrepancy between the systems' effectiveness and participants' perceptions. Participants rated the system in terms of how well they thought they had learned with the developed AI system significantly lower than usability ratings, highlighting that while the system was effective at providing immediate accessible information, it may have been limited in providing sustained knowledge retention. Overall, this research identified the trade-offs between accessibility and learner engagement that need to be considered when developing similar applications.

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1.Introduction

The development of Artificial Intelligence (AI) has advanced the landscape of education by providing individuals with personalised, adaptive environments which can enhance learning outcomes (Fortuna et al., 2025). AI-driven learning has been shown to significantly improve learners' engagement, motivation and retention, which is important because student engagement is considered to play a crucial role in supporting academic success (Alhassan et al., 2024; Zaharuddin et al., 2024). This is particularly significant because research demonstrates that students who learn in traditional instructional environments are 1.5 times more likely to fail than students who are in active learning environments (Freeman et al., 2014).

Despite this, astronomy persists to present significant pedagogical challenges, as learners often face difficulties when conceptualising abstract astronomical phenomena (Lelliott and Rollnick, 2010). In comparison to many other scientific disciplines, where learners can observe or experiment with concepts (National Research Council, 2000, pp.12–13, 23), astronomy requires visualisation of spatial relationships, planetary motion, scale, direction and distances (Cole et al., 2018), which can often be difficult to understand, as they cannot be directly observed (Ozkan and Akcay, 2016). As a result, individuals often hold a high prevalence of persistent misconceptions that are particularly difficult to correct only using traditional instructional approaches (Lelliott and Rollnick, 2010; Guerra-Reyes et al., 2024). While there has been extensive research exploring how AI-driven applications and intelligent tutoring systems influence learning (Ma et al., 2014; Wang et al., 2024), there has been very limited research regarding these systems in an educational astronomy context (Okulu, 2025; Díaz, 2025).

To address this, this research aims to investigate the effectiveness of a voice-interactive AI-driven learning platform compared to traditional instructional approaches in enhancing learning outcomes in astronomy. A React astronomy application was built and then evaluated through participant investigations, which assessed quantitative and qualitative results. The project's scope included developing, testing and then investigating the system and the planned limitations include a small sample size and a limited testing time.

2. Research and Planning

2.1 Problem Definition and Research Aims

Textbooks in astronomy often create further problems for students, with research suggesting that textbook images do not always support student understanding and can create further confusion (Testa et al., 2014), whereas AI-driven learning systems can provide a more adaptive and personalised experience (Gutierrez and Atkinson, 2011; Naseer and Khawaja, 2025). Therefore, the research question explored in this study was: “*Can a voice-interactive AI-driven astronomy learning platform produce greater learning gains than traditional e-textbook methodologies in astronomy?*”.

Additionally, it was hypothesised that:

H₁ The developed AI system will have significantly greater learning outcomes than the e-textbook in the post-study assessments.

H₀ There will be no significant difference in learning outcomes between the two methodologies in the post-study assessments.

2.2 Literature Review

2.2.1 Overview

As previously discussed, astronomy presents educational challenges due to requiring the learner to understand abstract astronomical subjects (Lelliott and Rollnick, 2010). Regardless of these challenges, several technological solutions have been developed. For example, Intelligent Tutoring Systems and AI-based learning systems can provide adaptive, personalised learning, which can support understanding by providing directed feedback (Gutierrez and Atkinson, 2011; Naseer and Khawaja, 2025).

Therefore, this review investigates the role of these instructional approaches in addressing learners' misconceptions in astronomy, with a strong emphasis on evaluating learning outcomes and the effectiveness of these instructional approaches.

2.2.2 Misconceptions in Astronomy

Existing literature suggests that misconceptions are deeply rooted throughout the course of study. Gali (2021) explored 38 students' misconceptions of basic scientific concepts and found that students held multiple misunderstandings that remained persistent after instruction.

This suggests that misconceptions may not be easily corrected by traditional approaches, highlighting that conceptual understanding may be hindered.

Furthermore, Kanli (2014) identified that the issue extends further than students, as teachers themselves may hold incorrect beliefs about basic astronomical concepts. Using a modified three-tier ACAT questionnaire, Kanli (2014) tested misconceptions on 388 pre-service and in-service teachers. The study identified that instructors held incorrect beliefs on fundamental topics, such as lunar phases, the sun's positioning and the causes of seasons. As a collective, this suggests that traditional instructional methodologies may not be sufficient for addressing learners' misconceptions and in some cases, they may continue to persist after instruction (Duit and Treagust, 2003; Gali, 2021).

These traditional methodologies have been consistently critiqued by literature, with textbooks being criticised for inaccurate spatial sizes and distances (Taylor and Grundstrom, 2011). This can be particularly problematic, as astronomy requires the learner to visualise spatial relationships and interpret them correctly to form an accurate mental model (Cole et al., 2018). As a result, these textbook representations may in fact lead to learners forming misconceptions. For example, Gorecek Baybars et al. (2018) highlight that textbook diagrams may contribute to learners believing that stars in a constellation appear to be lined at similar distances. This is further reinforced when Kiroğlu (2015) argues that concepts within astronomy are difficult to teach without 3D modelling, suggesting that textbooks may be limited in supporting accurate spatial understanding. Overall, this further highlights the need for instructional approaches that can directly address learners' misconceptions.

To overcome these challenges, research has investigated approaches that can confront learners' misunderstandings in astronomy. For example, Diakidoy and Kendeou (2001) investigated the learning outcomes of 63 students by comparing instructional approaches that tackle learners' misconceptions with traditional textbook methodologies, whereas Zeilik et al. (1997) implemented a "conceptually based instructional model" designed to target students' misunderstandings. Both studies found consistent results, as conceptual understanding significantly improved when the implemented approach explicitly confronted these misconceptions. As a result, this outcome supports Conceptual Change Theory, which suggests that for learners to overcome their misconceptions, they must first identify, challenge and restructure their existing incorrect mental models (Posner et al., 1982). This

highlights the importance of instructional approaches that can facilitate conceptual change within learners.

However, while these studies explain how to address misconceptions, they remain limited in providing any technological solutions, highlighting a potential research gap for adaptive learning systems that can support education in astronomy.

2.2.3 Intelligent Tutoring Systems (ITS)

Intelligent Tutoring Systems may be an effective tool at challenging learners' misconceptions in astronomy, as they can provide individualised learning support by adaptively responding to learners' cognitive profiles (Zerkouk et al., 2025), through providing dynamic feedback, personalised hints and explanations (Kochmar et al., 2020).

Existing literature presents a diverse collection of results regarding the effectiveness of Intelligent Tutoring Systems. Steenbergen-Hu and Cooper (2014) conducted a meta-analysis consisting of 39 studies and found that ITS relatively improved students' learning outcomes ($g=0.32-0.37$), suggesting that ITS can provide instructional benefits over other traditional methodologies. However, earlier research investigating answer-based, step-based and sub-step-based ITS, reported a significantly higher effectiveness ($d=0.76$) comparable to the learning outcomes of human-based tutoring ($d=0.79$) (VanLehn, 2011).

The variation in results highlights a key inconsistency in the findings of ITS, as it demonstrates that the findings may vary depending on the research context. This implies that, in the investigation, learning outcomes may be impacted by a variety of external influences such as system design, educational setting and participant characteristics (Schneider and Simonsmeier, 2025), highlighting the gap for more consistent implementations that produce reliable results across all instructional environments. This inconsistency is further reinforced by Bloom (1984), documenting that one-to-one tutoring had an effect size of 2σ , suggesting that although ITS produced favourable learning outcomes, they may still be less effective in comparison to human-based approaches.

This limitation is directly addressed within a more extensive meta-analysis comparing the effectiveness of ITS and non-ITS learning environments. Ma et al. (2014) analysed a substantial amount of effect sizes (107 effect sizes), derived from a wide range of contexts, suggesting it has the potential of providing stronger, generalisable evidence to support the effectiveness of ITS. The findings demonstrate that learning outcomes improved compared to traditional textbook methodologies, with results comparable to Steenbergen-Hu and Cooper's

(2014) research, suggesting a consistency between findings. This is significant because it helps to mitigate the variation observed in the previous results, highlighting that ITS can be a moderately effective tool.

However, while these studies show that ITS can improve learning outcomes, the variation in findings suggests that there may be a need for instructional methodologies that produce more consistent results. Additionally, they are limited in providing solutions specifically in an astronomy educational setting.

2.2.4 Modern AI-based Learning Systems

Extending on the findings of ITS, recent research has explored the usage of AI learning systems. Multiple studies employed a randomised controlled trial (RCT) to investigate the effectiveness of AI learning systems in comparison to other instructional strategies. Kestin et al. (2025) analysed the learning outcomes of 194 students using an AI chatbot compared to an active learning class, whereas LearnLM Team Google et al. (2025) provided 165 students access to an AI-generative model (LearnLM). There were consistent results between the studies, with Kestin et al. (2025) finding that students who were using the AI chatbot learned information at a faster rate than an active learning classes, and LearnLM Team Google et al. (2025) reporting that LearnLM produced similar outcomes to human-based approaches. Overall, these results indicate that AI-driven learning approaches can provide learners with an effective tool for learning.

However, it is important to acknowledge the limitations of both studies. While Kestin et al.'s (2025) research was conducted in a controlled environment, which could have limited its generalisability to a wider educational context, LearnLM Team Google et al.'s (2025) method included human tutors being able to supervise AI's responses and having the ability to edit them. This further limited the study, as in a real-world scenario, constant human supervision may not be feasible. Additionally, it may have also inflated the AI's measured outcomes, as it could have improved the response accuracy further than what the system could provide.

In contrast, Bastani et al. (2025) conducted a study which analysed 1000 students using an unstructured standard ChatGPT interface (GPT-Base) compared to an interface with structured guardrails implemented (GPT-Tutor). The findings demonstrate that although AI improved short-term learning outcomes, students' performance significantly decreased when AI was removed. A potential reason for this could be that students were over reliant on the generated AI responses to answer the questions, which may have reduced the cognitive

engagement with the system. As a result, this could have inflated the short-term learning outcomes observed. However, it was further noted that these negative effects were mitigated when students studied using the structured AI system, highlighting that the system's design is important when it comes to supporting sustained learning outcomes. These findings are consistent with Kestin et al.'s (2025) research, which suggests that AI tutoring systems need to be "carefully designed" to improve learning outcomes. This highlights a key contradiction between the system's short-term performance and long-term learning outcomes as it's demonstrated that unstructured systems can provide effective immediate assistance but not sustained retention.

However, while this body of research highlights how the design of an AI system is important in affecting the observed learning outcomes, it is limited in providing alternate design solutions, such as voice-interactive systems, that may support learners' sustained outcomes.

2.2.5 Voice User Interfaces (VUIs)

An emerging AI-driven technology that supports user interaction is Voice User Interfaces (VUIs) (Zhou and Zhao, 2025). Existing literature investigates the usability of Voice User Interfaces by evaluating factors such as user experience and accessibility. Pyae and Joelsson (2018) conducted research comprising of 114 participants from a variety of geographical backgrounds and found that participants viewed VUIs as user-friendly, efficient and supportive in everyday activities. This outcome suggests that VUIs could have the potential in supporting learners through accessible interaction.

However, these findings should be interpreted with caution, as several limitations were identified. Firstly, participants reported that non-English words were not recognised correctly by the system, highlighting the potential low feasibility to an international context. It was further noted that the need to repeat commands, background noise and multiple voice inputs could reduce the system's performance, suggesting that while ease-of-use was a key positive of the system, the effectiveness may depend on the environment of where the system is located. Furthermore, the limited number of non-English speakers in the sample size reduces the generalisability to a wider international audience.

Further research has examined the use of VUIs in an older age learning environment. Desai et al. (2023) conducted a mixed-method study consisting of 10 older adults to evaluate the use of an interactive storytelling VUI. The study found similar results to Pyae and Joelsson (2018), highlighting that participants reported a generally positive learning experience, good

usability and engagement. However, their findings contributed further by identifying that control, credibility and compassion were essential factors for the participant's potential sustained use. This expands on the prior studies as it demonstrates that while a system can be easy-to-use, it doesn't necessarily contribute to continued usage. However, the small sample size limits the reliability and generalisability of these findings.

Overall, while VUIs demonstrate that they can be effective in providing an easy-to-use and engaging experience, the previous research is limited in exploring how these systems can integrate with a visual learning environment, such as augmented reality experiences (Pyae and Joelsson, 2018; Desai et al., 2023).

2.2.6 Augmented Reality Visual Learning Environments

Augmented Reality (AR) is a technology that can provide learners with enhanced spatial understanding by enabling interaction with three-dimensional objects (Chow and Nazlee Sharmin, 2025). Previous studies have investigated how augmented reality visual environments can support learning in astronomy. Fleck and Simon (2013) studied the learning outcomes of 39 young students using an AR model compared to a physical model. The study demonstrated that learners were more likely to develop scientific conceptual understanding while using the AR model. This suggests that AR can provide a more effective learning experience for developing deeper conceptual understanding than physical models within this investigation's context. Similarly, Ferrari et al. (2024) found that 130 students using an AR-based application for learning astronomy showed a considerable increase in academic performance. These consistent findings suggest there is strong evidence that learning outcomes can improve by allowing students to interact, visualise and simulate virtual celestial bodies, which can support learners as they can test their own theories. It was further noted that AR can provide a more engaging experience for students than traditional instructional methodologies, highlighting the importance of AR technologies in supporting effective learning for students.

However, there were several limitations within both studies. Firstly, the sample of both studies only included younger-aged students, which potentially limits the generalisability of the findings to an older adult audience. Additionally, Fleck and Simon's (2013) research primarily focused on a narrow collection of astronomy topics, which may have limited the findings' reliability when considering other concepts in astronomy.

2.2.7 Research Gaps and Justification for the Study

Although the existing research in this literature review highlights the potential educational learning benefits of systems such as intelligent tutoring systems, AI learning systems, voice user interfaces and augmented reality environments (Ma et al., 2014; Kestin et al., 2025; Pyae and Joelsson, 2018; Ferrari et al., 2024), there remains limited research exploring how integrating these functionalities into one single system may enhance learning outcomes (Xie et al., 2025).

Therefore, this project addresses this research gap by developing a system that integrates voice-interactive AI-based learning into a visual three-dimensional environment. The study will then explore if this approach enhances learning gains greater than what traditional methodologies, in this case, e-textbooks, can provide. In developing this system, this study aims to provide a more comprehensive understanding on how an integrated approach can support learners' deeper conceptual understanding within astronomy.

2.3 Design Approaches

As the literature review identified a clear research gap, there needs to be an evaluation of the possible techniques that can be implemented to meet the research aim. This is essential in identifying the most appropriate approach for promoting effective astronomy education within this study.

Implementing an existing third-party AI chatbot interface was considered, as it would significantly reduce the system development time. Existing literature replicated a structured and unstructured ChatGPT interface to evaluate the learning outcomes of students. It was found that while students were completing tasks with the AI interface, their performance significantly improved. This outcome demonstrates the potential effectiveness of using a third-party AI chatbot within a comparative study (Bastani et al., 2025).

However, this approach presented significant drawbacks for the investigation. Firstly, the literature review identified that there was a research gap of integrating multiple components, such as voice-interactivity, AI learning and three-dimensional environments, into one single astronomy learning system. Existing AI chatbot interfaces typically comprise of text-based interaction and often do not provide any additional functionality or visual environments, which would address this gap. Therefore, this approach was unable to support the learners' spatial reasoning, which was identified as an important factor in understanding visual concepts (Höffler, 2010). As a result, this approach was deemed less suitable, as it would not

sufficiently support learners' conceptual understanding within the investigation phase, which was essential to meet the research aim.

Alternatively, a custom-built astronomy learning platform was selected to be developed for the investigation phase. Therefore, this study adopted a hybrid approach of development and experimental research. This allowed for the system to be designed in accordance with the literature review's research gap by enabling the integration of multiple components, such as voice-interactivity, AI learning and spatial environments. However, this approach significantly increases development time and complexity, which is a significant drawback due to the project's limited scope. Nevertheless, this approach was deemed the most suitable, as it enabled a more controlled environment, essential for comparative analysis.

2.4 Research Approaches

As the selected approach was to implement a custom-built astronomy learning platform, there were multiple research strategies that could be utilised to evaluate the effectiveness of the system.

An observational study approach was considered for the investigation phase. Previous research has employed this approach to identify pedagogical strategies utilised by clinical teachers (Nilsson et al., 2010). Nevertheless, while this methodology can be used to capture real-world behaviours, it remains limited in providing quantifiable empirical evidence to support learning outcomes. Therefore, this methodology was not suitable for this study, as the primary research aim was to compare numerical learning outcomes between instructional approaches.

Surveys and interviews were further considered as a primary research approach, as they can be utilised to capture user experiences, perceptions and behaviours. Previous research employed this methodology to investigate augmented reality uses in astronomy education. The study found significant findings regarding user engagement and perceptions. However, it was further noted that direct evidence for learning outcomes remained limited, an important metric needed for comparative analysis within this study (Rafał Kaźmierczak et al., 2025). Additionally, this approach relies on subjective self-reported data from participants rather than providing measurable learning outcome data, which in this investigation's context may reduce the validity of the measured findings. Therefore, this approach was incompatible as it did not meet the research objectives of this study.

A quasi-experimental approach was also considered to evaluate learning outcomes. Previous research has utilised this approach to analyse conceptual understanding in astronomy by assigning two existing classes into an experimental and control group (Sharp, 2002).

However, this may introduce selection bias, as the differences in participants' characteristics are not randomly distributed between groups, potentially reducing the internal validity of the findings. Therefore, this approach was not suitable for this study as learning outcomes may be influenced by the participants' characteristics.

As a result, a randomised controlled trial (RCT) methodology was selected as the primary research approach within this study. Unlike the previous methods, this approach enables participants' characteristics to be randomly distributed across groups, overall reducing selection bias and improving the internal validity of the findings (Saldanha et al., 2022). Furthermore, using a crossover approach allows for the participants to experience both conditions, increasing the amount of data collected. However, this approach may introduce potential learning effects, where participants' previous exposure to the material influences the subsequent assessments. Despite this, the RCT crossover methodology was deemed the most appropriate approach for robustness within the investigation phase and fairness while evaluating learning outcomes.

2.5 Requirements Capture

Due to the hybrid nature of this investigation, it is important that the developed AI system must adhere to a list of system requirements that ensure the investigation process is streamlined. The system requirements were formulated through key conclusion found within the literature review, research into user behaviours and careful examination of alternative approaches.

2.6.1 Functional Requirements

FR1 - “The system should use voice-interaction as the main source of teaching output.”

The literature review highlighted that from a Human Computer Interaction (HCI) perspective, users have reported that Voice User Interfaces (VUIs) can be efficient and easy to use.

However, it was further noted that there were several limitations, such as reduced accuracy for non-English words, background noise and multiple inputs (Pyae and Joelsson, 2018).

Despite this, voice interaction will be prioritised to promote user-friendly interaction during the investigation phase, at a cost of reduced response accuracy.

FR2 - “The system should address student misconception by scaffolding learning, asking questions and providing feedback”

One main challenge identified in astronomy is that learners often misunderstand fundamental concepts, due to the spatial reasoning required to understand abstract astronomical phenomena (Sneider et al., 2011). Intelligent Tutoring Systems can provide personalised feedback and scaffolding, which can assist learners in comprehending concepts (VanLehn, 2011). Therefore, this project will include these features to support learners’ conceptual understanding within the investigation phase.

FR3 - “The system should have structured guardrails in place to ensure that accurate and consistent astronomy subject is being taught.”

Text-based AI systems can generate responses that are either hallucinated, inaccurate, or off topic (Maynez et al., 2020), which would result in an unfair comparison during the investigation phase. Therefore, structured guardrails must be in place to ensure that responses remain accurate and aligned with the astronomy subject.

2.6.2 Non-functional Requirements

NFR1 - “The system should aim to provide accurate, consistent and reliable responses to ensure a fair comparison during the investigation.”

This requirement aims to ensure that the AI system maintains reliable performance for all participants during the investigation. Inconsistent outputs may reduce the fairness when evaluating the developed system against textbook methodologies. Therefore, the system should aim to provide accurate responses to help increase the results validity.

NFR2 - “The system should be engaging and easy-to-use for all users.”

It was found that some studies suggest that student engagement enhances learning (Parsons and Taylor, 2011). For this reason, the system should be engaging for all users during the investigation phase.

Due to this project’s limited scope, further requirements are included in Appendix A

2.6 Tools and Techniques

The system’s requirements provided key insights into which tools and techniques should be utilised to develop an effective learning tool for the investigation phase.

React was selected for the frontend of the system due to its component-based architecture, which enabled the system's user interface to be modular and easily scaled. React Three Fiber was chosen as it allowed a three-dimensional astronomy learning environment to be integrated into the system, which met the spatial needs of learners identified in the literature review. Furthermore, Node.js and Express were selected to securely manage session creation with the OpenAI API, ensuring that the API key never got exposed to the frontend (Meta Open Source, 2025).

Additionally, for the investigation phase, baseline test and post-test assessments were selected to measure the learning outcomes of both methodologies, and feedback forms were chosen to capture participants' perceptions. Microsoft Excel was selected over RStudio, as it was beneficial for efficient statistical analysis of small datasets due to its low complexity.

3. Practical Work

3.1 Hybrid Design

As a result of the research and planning stage, it was found that a hybrid approach of development and then experimental research was the most effective solution for providing measurable learning outcomes. Therefore, this project utilised this approach to investigate the effectiveness of a developed voice-interactive astronomy tutoring system compared to traditional textbook methodologies.

The study followed two distinct phases:

1. The development phase, where the AI system was built.
2. The Investigative phase, where the developed system was compared to traditional textbook methodologies.

The hybrid approach was beneficial in developing a system that addressed the discussed research gap in the literature review. It allowed for the integration of multiple features such as voice interactivity, 3D visualisation, and AI tutoring, whilst being able have its effectiveness evaluated against traditional learning methodologies.

Part 1 – Development of the Astronomy Application

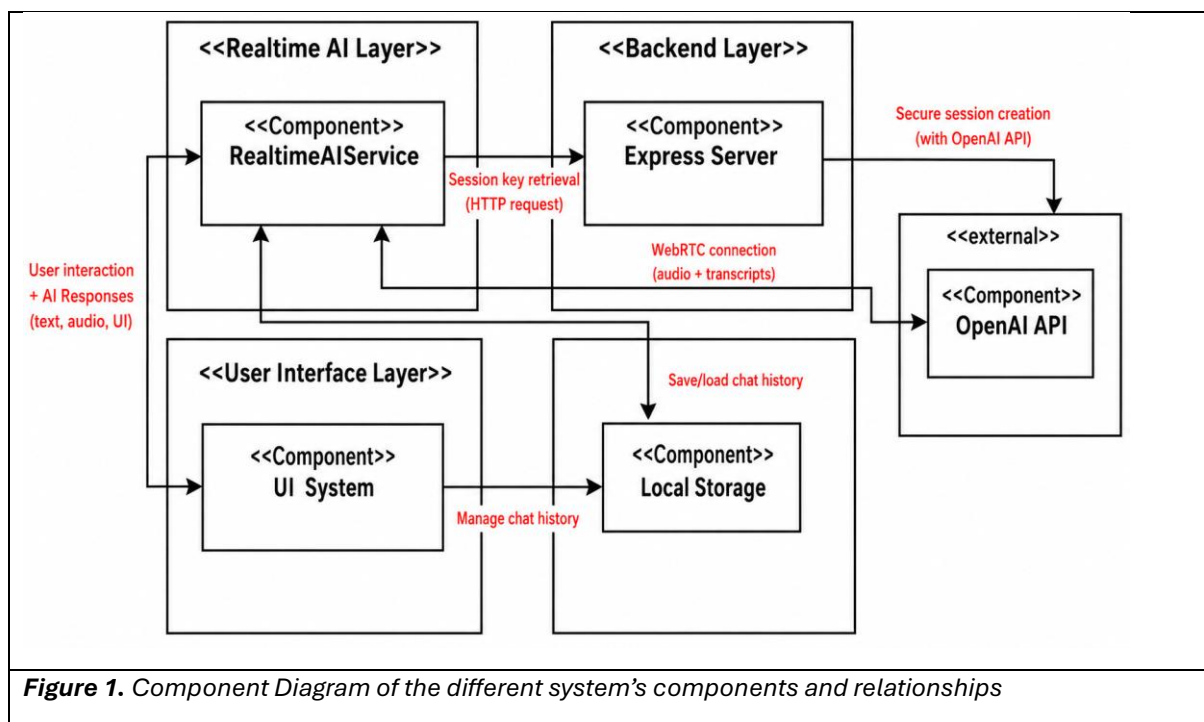
3.2 System Design

3.2.1 System Architecture

The system employed a client-server architecture with peer-to-peer (P2P) communication with the AI tutor. A monolithic approach was considered for its low complexity. However, modularising the system's architecture into three different layers: 1) user interface layer 2) real-time AI layer 3) backend layer, was designed to ensure a separation of concerns. This enabled the system's different responsibilities to be isolated throughout the architecture, at the cost of higher complexity.

The user interface layer handles the user's interaction within the system, enabling the users to interact with the different system elements such as the 3D environment, UI, and visual elements. The real-time AI layer manages real-time communication with the OpenAI API using WebRTC, and the backend layer handles secure session creation with the API.

As shown below, *Figure 1* illustrates the system's architecture using a component diagram. The diagram highlights the different system components and their relationships.



3.2.2 User Interface Layer

React was selected for the user interface layer due to its component-based design. This was advantageous compared to a vanilla JavaScript approach, as it enabled the system's UI elements to be split into reusable components, resulting in the system's maintainability and scalability significantly improving, at a cost of complexity. Additionally, React's state-driven rendering allowed for the system's UI elements, such as menus and buttons, to be easily integrated into the three-dimensional environment. This ensured that the application would enhance the overall user experience within the investigation phase and support the system's engagement requirements (Meta Open Source, 2025; react.dev, n.d.).

React's component-based design relied on a variety of external libraries, which increased the architectural complexity of the system. However, it was still considered an appropriate approach as its declarative architecture significantly improved efficiency whilst updating the system's UI elements. In comparison, vanilla JavaScript's imperative architecture would have required manual DOM manipulation, which may have reduced the maintainability in this implementation (Meta Open Source, n.d.; Mozilla Foundation, n.d.).

React Three Fiber was adopted in the user interface layer, as it could render three-dimensional scenes using three.js (*Pmnd.rs*, 2021). Three-dimensional environments provide increased support for learners' spatial understanding when compared to two-dimensional illustrations (Höffler, 2010). Therefore, R3F was pivotal in producing an interactive three-dimensional space environment for the investigation phase, which overall supported the abstract spatial needs of learners highlighted in the requirements.

Although dedicated game engine alternatives such as Unity may be able to provide improved graphical and rendering performance, R3F was selected due to its ability to be integrated within React's component-based architecture. Compared to Unity, which requires a canvas-based UI system, the R3F approach closely aligned with the user interaction requirements, as it enabled a loosely coupled UI design that allowed elements to be layered over the three-dimensional scene.

Moreover, R3F's modular design allowed the three-dimensional components, such as star fields, planets and animations, to be developed separately and then dynamically loaded into the scene when required. This significantly improved development overhead and allowed for the animations when travelling between planets to be fast and responsive, which was essential for an efficient investigation process.

3.2.3 Interactive User Interface (UI) design

As the system's primary focus was to create an immersive learning environment for participants within the investigation phase, a minimalistic design approach was adopted. Displaying fewer elements was intended to reduce distractions and support interaction with the main astronomy features. As a result, this closely aligned with the systems engagement requirements, as it enabled users to remain focused whilst completing the investigation's assessments.

Figure 2 highlights the minimal UI design that the system adopted. Planet textures were sourced from a public bitmap dataset by Hastings-Trew (n.d.) and applied to a sphere object. This ensured that the planet textures were realistic and suitable for the project. The system's central focus lies on the planet, as it acts as the primary interaction tool. Users can orbit, zoom and move around the planetary environment to gain a greater spatial understanding. Additionally, hovering over the planet will provide a randomised fact. This was implemented to provide users with greater control over how they want to progress and learn throughout the system, which helped support user engagement with the astronomy content.

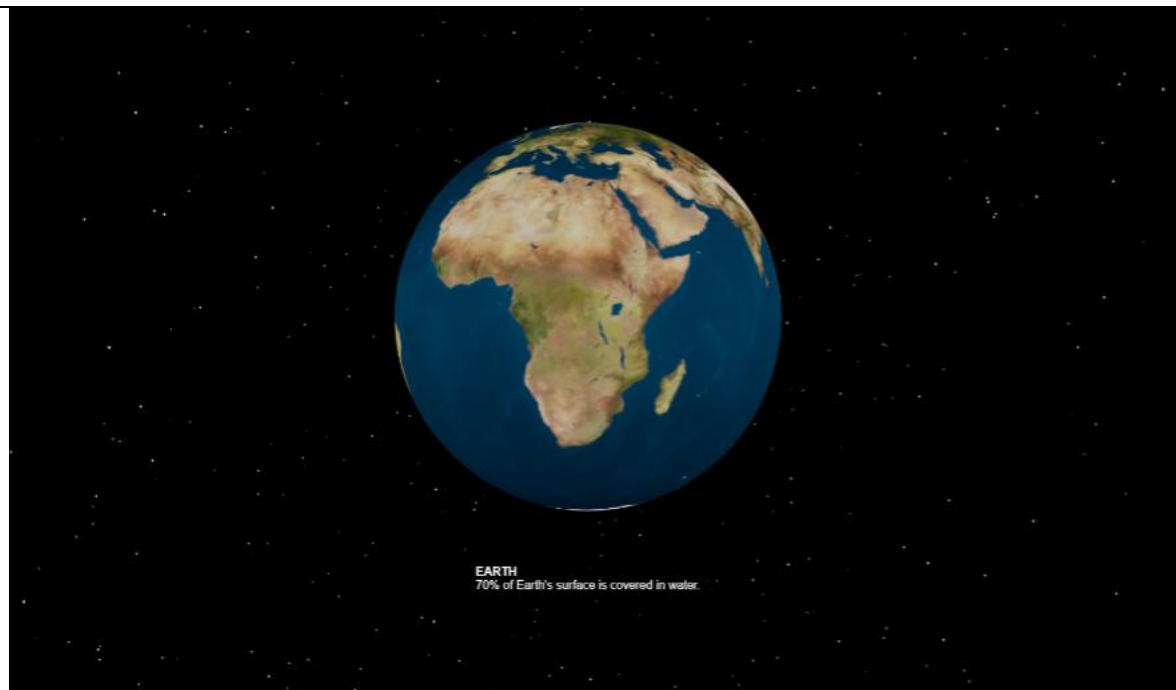


Figure 2. Screenshot Showing the Minimalistic Design Approach

Furthermore, subtitles were added to the system's UI to provide additional accessibility functionality. The subtitles would appear when the chatbot responds to the user, which

supported the engagement requirements by enhancing comprehension of the astronomy material (Gernsbacher, 2015). The subtitles of the system are shown in Figure 3 below:

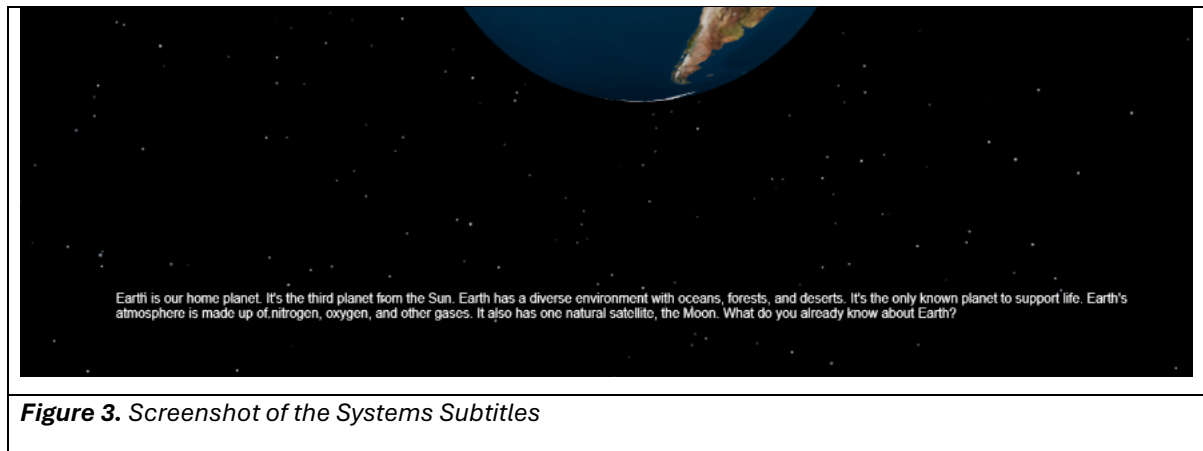


Figure 3. Screenshot of the Systems Subtitles

3.2.4 Real-time AI layer (WebRTC) and API Design

To create the study's astronomy AI tutor, the real-time AI layer utilised an OpenAI API (API) using Web Real-time communication (WebRTC). This was effective for enabling real-time voice interaction with the AI tutor, a requirement essential for natural learning within the investigation phase. The inclusion of an AI tutor was justified by previous research that demonstrates that intelligent tutoring systems can increase learning outcomes of students (Kulik and Fletcher, 2016).

While developing the AI model from scratch would have provided greater control over the AI's responses, this project's time constraints and the technical complexity would have significantly limited the effectiveness of this approach. As a result, an API-based design was utilised as it could generate higher-quality responses that could be integrated within the custom-built space application, whilst significantly reducing development time.

Additionally, the API followed a separation of concerns design principle, as it allowed for the AI model to be separately managed by OpenAI. This enabled the API to provide additional functionality such as handling context-driven responses, which further supported the system's requirements of scaffolding learning and providing feedback. Furthermore, the API supported minimal data retention of participants within the investigation phase, which aligned with the ethical considerations of this study (OpenAI, 2023).

The AI model used for the tutor was gpt-4o-mini-realtime-preview. Alternate models provided additional functionality such as image generation, transcription and speech generation from text (OpenAI, 2025). However, conversational agents have shown to

enhance user engagement and personalised learning within an educational setting (Alfehaid and Hammami, 2023), which suggests that the best potential approach to meet the project's requirements requires a real-time audio communication layer between the user and system.

A higher-performing gpt-realtime-1.5 model was considered. However, the mini-real-time model was selected as it provided adequate performance whilst being the most cost-effective. This ensured that the system was suitable for frequent use within the investigation phase. Additionally, the model provided faster response times than the higher-performing model, which was essential in supporting natural-flowing communication within the system (OpenAI, 2025).

The real-time model supported the use of either WebRTC or WebSockets. WebSockets provided a client-to-server communication architecture, which would have been less complex to develop in this implementation. However, a WebRTC approach was selected as it enabled a low-latency, peer-to-peer (P2P) architecture (Mahmoud and Raouf Abozariba, 2024) that allowed for real-time communication within the system. This approach was selected as it ensured that the AI tutor could be fast and responsive, a requirement essential to enhance the overall user experience within the investigation phase.

3.2.5 Backend Layer

The backend layer was developed using Node.js Express, which enabled the system to create a secure session with the OpenAI API. While this approach created significant backend complexity, it was essential in improving the security of the system by ensuring that the API key was never exposed to the frontend, only the temporary key. Additionally, the server provided further functionality to select a chosen model, voice and pass in certain instructions, which was fundamental in creating the controlled guardrails which were needed within the investigation phase. Overall, the implemented guardrails ensured that the responses from the AI tutor were accurate and aligned with the astronomy subject within the study.

See Appendix G for system prototypes and documentation (readme file)

3.3 Implementation

3.3.1 WebRTC Peer Connection and Data Channels

The `connectAI` function (see Appendix H) creates a WebRTC peer-to-peer connection between the client and the OpenAI API service. The process is handled by retrieving a

temporary authentication key from the backend, which is then used during the SDP handshake to authenticate the connection.

An `RTCPeerConnection` object is created (see Appendix H) to handle real-time audio streaming and data flow between the client and the API service. The audio is streamed to the API via a media track, and a data channel is responsible for handling the flow of transcripts. The audio and transcripts were separated to allow for the subtitles to be updated independently of the audio, improving the responsiveness of the system within the investigation phase.

A message event handler `dc.onmessage` processes data that is sent from the API, enabling the transcripts to be incrementally added to the system's subtitles. Once a response is fully completed, the entire transcript is sent to a `detectPlanetMention` function, which detects mentioned planets within the transcript and triggers an appropriate animation.

3.3.2 Planet Detection Algorithm and Transitions

A planet detection algorithm `detectPlanetMention` processes the transcript data received from the API to detect whether a planet was mentioned within the transcript. Prior to identification, the transcript was converted into lowercase. This enabled the system to identify a successful match consistently, without having to handle variations in capitalisation. The function then compares strings to see whether a planet name from the array `planetNames` is detected within the transcript. If any of the planets are detected, the system triggers an animation function `startPlanetTransition`, which animates to the detected planet. Animations are only triggered if the `currentPlanet` does not match the detected planet and if the transition is not already initialised, ensuring that animations or planet textures cannot overlap. (see Appendix H)

3.3.3 Local Storage

Local Storage is utilised to save and load users' chats within the system. On application load, the system retrieves previously stored chats and displays them in the chat history (see Appendix H). When a user converses with the AI, the new responses are appended to the chat history and saved to Local Storage, enabling users to review previous interactions made within the system. Previous chats can be removed from the chat history and Local Storage, enabling users to have control over what is stored within the application's history.

This was essential to allow participants to revisit previous chats whilst conducting the assessments, allowing for enhanced accessibility to information. All participants' chats were deleted from the chat history after the investigation phase was completed to stay in accordance with the ethical considerations.

3.4 Testing

Before the participants took the study, it was essential that the system was tested to ensure that all the core functionalities would work as expected. This was vital for the investigation process to run smoothly and for the gathered results to be reliable and valid. Considering that the user flow throughout the system is relatively simple, the implemented testing strategy primarily focused on ensuring that the user experience would be consistent throughout the investigative process.

As a result, a black-box methodology was adopted as the primary testing strategy, allowing the system to be tested from the participant's perspective. While there was an argument that white-box testing could be utilised to identify logical errors from within the system, the investigation aimed to assess the effectiveness of the system in improving participants' learning outcomes rather than evaluating implementation quality. Additionally, as a large proportion of the AI functionality was handled by a third-party (OpenAI), it reduced the need for extensive white-box tests to be implemented.

Testing coverage focused on the user-interactive functionalities within the system, ensuring that all the vital system components could be validated before participants started the process. Coverage was achieved through techniques such as functional, integration, edge case and usability tests. Functional tests were conducted to verify that the system's individual features, such as the AI responses, UI updates, inputs and outputs all worked and would produce comprehensive results prior to the investigation. Integration tests were conducted to ensure that the system's different components would work together as intended. This was essential to the investigation, as it focused on assessing the system's effectiveness during user interaction. Edge case tests were conducted to assess the unexpected conditions that may occur within the investigation such as unclear and invalid inputs, loss of internet connectivity and unexpected background noise. Usability tests were conducted using a pilot participant, who completed a full test run of the investigation and provided feedback, ensuring that any system issues could be promptly resolved prior to the main investigation process.

The selected testing strategies were sufficient for this investigation, as they primarily focused on ensuring that all the user-interactive functionalities were validated before the investigation. The implementation of functional, integration, edge case and usability tests were effective in identifying if the system contained specific issues such as hallucinated AI responses, UI inconsistencies and external influences. This was significant as it enabled the project to follow an iterative design flow to improve these features, which overall reduced the likelihood of participants experiencing technical issues within the investigation phase. As the testing strategy consisted of primarily black-box testing, it did not fully account for all logical errors that could be within the system that would be identified more thoroughly through white-box testing. However, the strategy still provided a strong confidence level for the system's robustness within the investigation phase.

See Appendix I for the full test documentation.

Part 2 – Investigation Comparing the Effectiveness of the Developed AI System and Traditional Textbook Methodologies

3.5 Investigation Design

3.5.1 Randomised Controlled Trial (RCT)

To compare learning outcomes between the developed AI system and traditional textbook methodologies, a Randomised Controlled Trial (RCT) crossover method was conducted on a small university student-based participant pool (n=8). Although the RCT framework was conducted in a highly controlled environment, which may have affected the participants' natural learning behaviours, this approach ensured that the participants studied using both conditions and were randomly assigned topics. The RCT framework also reduced selection bias through randomising participants into their initial conditions (Saldanha et al., 2022).

3.5.2 Baseline Assessments

The initial plan for the investigation design required the participants to take a baseline assessment without prior study, enabling learning outcomes to be calculated through comparison between the baseline and post-study assessment scores. To ensure that participants could not prepare for the assessment, the baseline assessment topics were not disclosed prior to the study. The assessment consisted of 20 astronomy questions with 10 questions on Topic A (Jupiter) and 10 questions on Topic B (Black holes). This was designed to keep the participants' experience consistent throughout the assessment, which limited the potential bias and ensured equal coverage between topics.

However, when it came to implementing the baseline assessment, it was further utilised for filtering participants based on whether they scored under or over a pre-defined threshold. This was implemented to mitigate against a potential ceiling effect, where high-scoring participants would limit the study's ability to differentiate between the high scores achieved and genuine learning gains. It is commonly viewed that a ceiling effect may exist if at least 15% of participants achieve the highest obtainable score on an assessment (Saithna and Cote, 2024). Therefore, participants with test scores under 19 marks were selected to undertake the study, compared to those whose test scores were greater than 19, being filtered out (1 mark less than the assessment's highest achievable score).

3.5.3 Participant Selection

Initially, other frameworks were considered for participant selection, such as allowing participants with any level of astronomy proficiency to take part in the study without completing a baseline assessment. However, this approach was rejected to minimise the influence that prior knowledge would have on post-test scores, ensuring that any test score differences represented the learning methodologies' effectiveness rather than initial subject knowledge.

As previously discussed, participants were selected based on their initial baseline results. In addition, the sample was further restricted to students between the ages of 18 and 60 to mitigate against several key ethical risks. The use of artificial intelligence within the investigation posed psychological, emotional and data privacy risks that could potentially harm vulnerable individuals outside of the study's accepted age range. Furthermore, the study required participants to interact with different system elements, which the accepted age range was expected to possess the required technical skills for assessment completion.

3.5.4 Experimental Conditions and Randomisation

The study's conditions consisted of two pedagogic groups: the control group, which studied using a searchable e-textbook (Fraknoi et al., 2022), acting as a traditional method and the experimental group, which used the developed AI system to study. The e-textbook was implemented to keep the comparison between methodologies fair and consistent, as both conditions were conducted on the same computer.

Participants were randomly assigned to the initial conditions using an 'out of the hat' selection technique. This is where participants had an equal chance of being drawn out of a hat into either of the conditions. Implementing this technique allowed the study to ensure that

the participants were genuinely in randomised groups, reducing the selection bias and improving the internal validity of the study.

3.5.5 Crossover Design

The study initially planned to use a between-subjects design, where participants only experienced one condition. However, to ensure that the differences in prior knowledge were balanced across conditions and due to the small sample size ($n=8$), a crossover design was selected as a more appropriate approach. The design allowed participants to experience both conditions within the study, effectively allowing for double the amount of data points to be collected on each participant. This allowed for the study to have a more informative dataset, which overall improved the reliability of the results collected.

3.5.6 Controlled Study Duration

Participants in both conditions experienced a controlled ten-minute study period. This ensured that participants could have equal amounts of exposure to the study content.

Initially, an alternate design was planned, where there was no time constraints implemented within the study period. However, this method was rejected as both conditions needed clear consistency between them. Without a fixed time constraint, it would have been more difficult to evaluate whether the learning outcomes were derived from the learning method or the differing study durations. As a result, the implemented time constraint ensured consistency, which enabled a fair comparison between the conditions.

3.5.7 Study Topic Selection

An alternate design approach that was considered was to only use a single astronomy topic for both conditions. While this design would have significantly reduced implementation complexity, there was a risk that already having exposure to the previous conditions study materials would have influenced the post-study assessments' performance on the subsequent condition.

Therefore, to reduce the carryover effects between conditions, two astronomy topics that are in different areas of physics were chosen for the study (Jupiter and Black holes). The topics were chosen as they were expected to have similar difficulties and to require the learner to use spatial reasoning. As a result, this helped ensure that any differences in post-study test performances were due to the effectiveness of the learning methodology rather than previous influences, enabling a fair comparison between conditions.

3.5.8 Assessment Design

To measure the effectiveness of each learning method, post-study assessments were conducted to evaluate participants' individual learning gains. The post-study assessment's structure consisted of ten astronomy questions on the topic that the participant had just studied during the condition. This was designed to measure the learning gains by calculating the difference between the baseline score and the post-study score. The post-study test questions were identical for both conditions, which ensured consistency throughout the assessments and allowed for the results to reflect the learning methodologies' effectiveness rather than differences in the questions asked.

Questions were structured as: 1) multiple-choice questions with one correct answer. 2) True/False questions. 3) open-ended questions. This approach was implemented to target participants' diverse types of thinking skills and to prevent the possibility of guessed correct answers influencing overall learning outcomes.

(An example of each type of assessment question is provided in Appendix E)

3.5.9 Assessments Marking Scheme

Post-study assessments were marked using a marking scheme supported by credible, high-quality resources. This ensured that there was a standardised correct answer for each question asked within the study. Objective test questions were marked using only one fixed answer. This enabled consistency throughout the marking of the assessments, which overall improved the reliability of the results gathered. Open-ended test questions were marked using sample answers derived from credible resources, resulting in a less complex marking process. However, as there was no fixed answer, the marking scheme required the marker to use their own judgement whether the question was right or wrong, which overall depending on the marker, could have affected the reliability of the results.

An alternative approach, utilising a second marker, was considered. This would have reduced the first marker's bias by ensuring that their judgments were accurate. However, this approach was rejected as it was not ensured that the marker would produce valid judgments for the study's questions. Additionally, the project's limited scope would have made it difficult to recruit a non-biased candidate for this role.

(An example of the marking scheme is provided in Appendix F)

3.5.10 Qualitative Feedback Forms

Qualitative feedback forms were implemented to collect users' perceptions of the learning methodologies. As the RCT design was the primary evaluation methodology, qualitative feedback was collected to complement the quantitative findings. This ensured that the results did not only reflect numeric learning outcomes but also a deeper understanding of participants' behaviours and perceptions during both learning methods. This provided further context into the meaning of the findings rather than just providing numerical data, which overall strengthened the credibility of the findings.

3.5.11 Sample Size and Recruitment

Participants primarily consisted of university students between the ages of 20 and 23, as they were considered the most appropriate population to interact with the system's elements. Recruitment was conducted via word of mouth. Initially, other recruitment methods were considered, such as advertising on social media. However, they were not implemented due to the project's limited scope.

The sample size for this study was $n=8$, which was sufficient for this project's limited time constraints. However, it may have limited the reliability and generalisability of the results.

3.5.12 Data Analysis

Quantitative data was analysed using a learning gains figure, which was calculated as the percentage points difference between the baseline and post-study test scores. Descriptive statistics were calculated to compare performance trends between both conditions. These included the mean, median, standard deviation and range.

An independent samples t-test was conducted to identify whether there was statistical significance between the AI and textbook conditions. Cohen's d was used to calculate the effect size, which measured the practical significance.

Qualitative feedback forms were collected and analysed through thematic analysis (Braun and Clarke, 2006). Different patterns were identified and grouped into themes, such as accessibility, knowledge retention and interactivity.

3.5.13 Ethical Considerations

This study's ethics were in accordance with Northumbria University and the Data Protection Act (2018). As this study required participants to interact with an AI-based large language model, additional considerations were in place to mitigate ethical risks. Prior to the

investigation, participants were notified via a *Participant information sheet* and a *system disclaimer* that the system may generate biased or inaccurate responses. Furthermore, the system development followed a pre-defined AI ethical considerations document, which promoted responsible, transparent AI usage throughout the investigation phase. (see Appendix C)

3.6 Investigation Procedure

Prior to beginning the study, participants were given a *Participant information sheet* (see Appendix D) outlining the important ethical considerations and procedures that were going to occur during the experiment. They were required to read through the document until they understood how the study was going to be conducted.

They were then informed that the study would take approximately 30 minutes to complete and that it was going to be conducted following a crossover framework. A further briefing was given explaining the two different astronomy subjects and learning methodologies going to be used within the study.

Proceeding this, a consent form was handed out which participant were required to sign prior to taking part, which ensured that everyone within the study had agreed voluntarily to participate. They were further informed that they could withdraw from the study at their own discretion.

Firstly, participants were required to take a baseline test consisting of twenty astronomy questions on Topic A and B before they could start the main comparative phase of the study. Before the baseline test commenced, it was highlighted that there were no time constraints and they could work at their own pace. Once participants had worked through the assessment, they were required to submit their answers for evaluation. Based on baseline performance, they were informed whether they had been selected for the comparative phase of the study.

The participants that were selected for the study were then shown how to use the learning materials in both conditions. An in-person tutorial explaining the fundamentals of how to navigate the learning method was given and any questions that needed to be answered were provided.

A question sheet was provided consisting of ten astronomy questions related to the first topic of study. Ten minutes were given to search and revise the answers to the questions on the question sheet using the assigned pedagogical methodology. Participants within the control

textbook condition studied using a provided e-textbook source, whereas participants in the experimental group interacted with the developed voice-interactive AI solution.

After the study periods ten minutes were completed, the question sheet and any notes made were removed and a post-study assessment was required to be taken. Participants were notified that there was no time constraints set for the completion of the assessment.

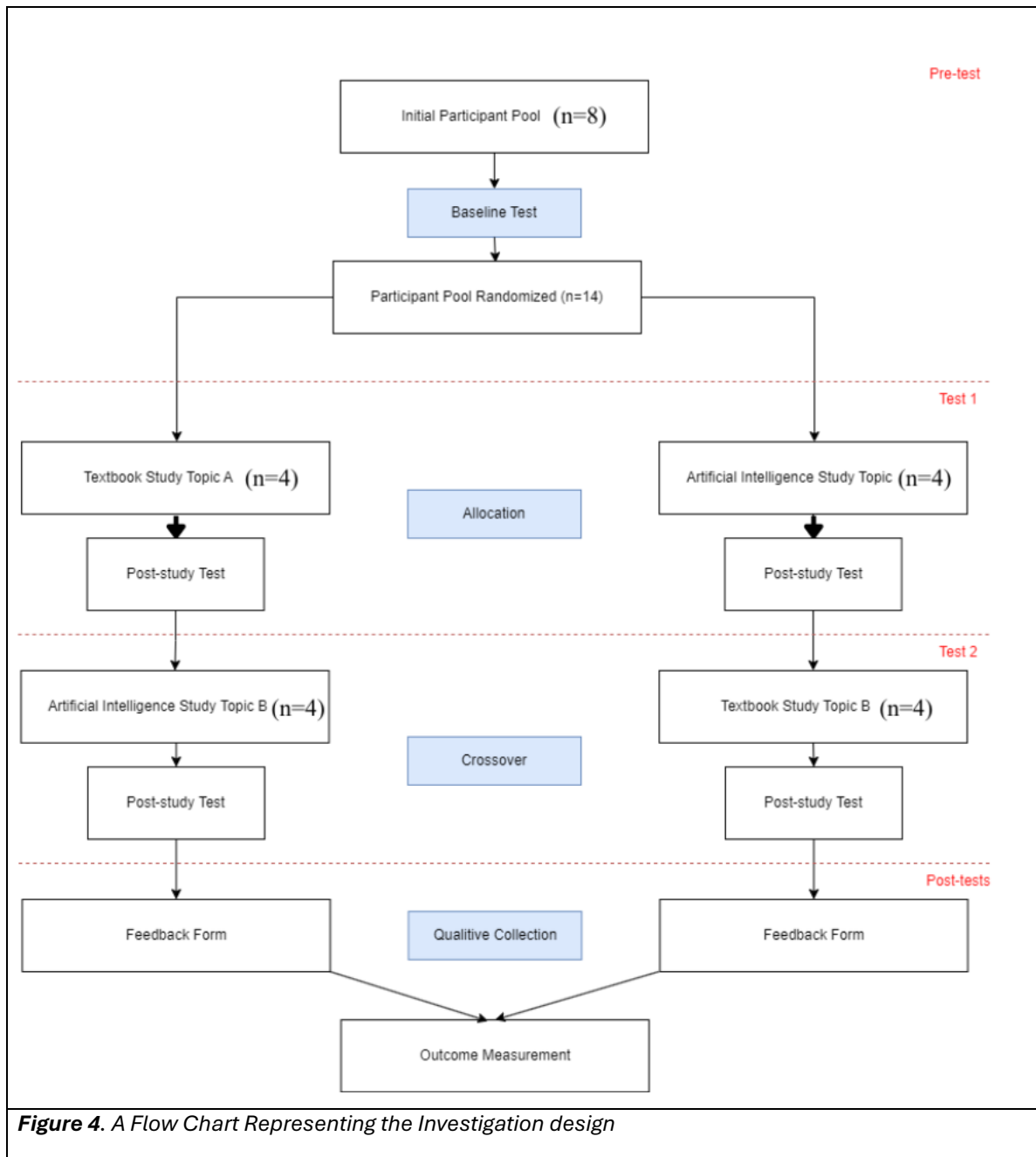
Following the assessment, the study followed a crossover approach. Participants were required to switch learning methodology and astronomy topic to the ones they had not previously experienced. The procedure then followed an identical structure to the initial completed conditions. Ten minutes was allocated to the study period to allow adequate study and revision of the material.

Proceeding the second study period, a post-study assessment was required to be taken. This followed all the procedures that the initial post-study assessment followed.

Once completed, the results from the baseline test, and the two post-study assessments were collected, stored and anonymised following the ethical standards outlined within the study. The assessments were then marked, and the data was analysed and comparatively evaluated.

Lastly, participants were asked to submit qualitative feedback regarding key factors within the study. Feedback points relating to the perceptions of the study material, and any further observations were required to be answered. Once feedback was collected, the data was anonymised and was used to refine the system for the next phase of candidates and to provide essential insights into the perceptions of the pedagogy used within the study.

Figure 4 presents a flow chart of the investigation procedure outlined *above*.



3.6.1 Assessment Environment and Data Collection

The investigation was conducted in a controlled environment within an unoccupied university classroom, specifically at Northumbria University. Participants completed the study independently, with investigations including two or more people either took at a distance or in separate classrooms to minimise external influence. All participants were provided with standardised instructions, and all procedures were kept identical to keep consistency throughout the study.

Data was collected through submitted online Microsoft Forms question sheets. The baseline assessments, post-study assessments and qualitative feedback forms were recorded, anonymised and stored securely in Microsoft OneDrive. Qualitative feedback regarding participants reflections on each learning method was collected at the end of the study.

3.6.2 Investigation Challenges

One challenge of the investigation was that some participants required extra guidance on navigating the learning material during the study period. This significantly reduced the amount of time that participants had to complete the remaining questions on the assessment. As a result, this reduced the fairness of the comparison, as several participant assessment performances may have been affected by the decreased time limit.

Another key challenge observed during the investigation was that there were variations in how long participants took to complete both condition's study period. Some participants in the AI condition completed the study period quicker than the allocated time and took the assessment early. Consequently, this affected the consistency between conditions and reduced comparison fairness.

Furthermore, it was observed that some participants using the AI condition would repeatedly ask the same question to the system without utilising the chat history function. This resulted in some participants requiring reminders that there was a chat history feature included within the system. As a result, this contributed to an external influence on participant interactions with the system, which affected the results accuracy.

3.6.3 Adaptations

Encountering these challenges during the investigation phase required several adaptations to keep the process streamlined. Prior to the assessment, participants requiring additional guidance were provided with further support. This included providing a structured in-person tutorial on how to navigate the system and contextual hints if the user became stuck. This was implemented to ensure that there were less interruptions during the investigation process, allowing participants to have consistency between both conditions. As a result, there was a fairer comparison between both groups.

Furthermore, participants who completed the assessment quicker than the allocated time were allowed to take the post-study test early. Although this affected the consistency between both conditions, it ensured that the participant engagement level was high. If this was not

implemented, then participants may have become distracted before taking the post-study test, which would have affected the reliability of the results.

Lastly, it was found that one of the baseline assessments questions contained factually incorrect information. This was corrected during the marking process by removing the incorrect question out of the final score. This resulted in the assessment's mark changing from being out of 10 to out of 9.

3.7 Results

3.7.1 Quantitative Results

3.7.1.1 Descriptive Statistics

An effective comparison between both learning methodologies required a learning gain benchmark to be calculated. Learning gains were calculated as the mean percentage points difference between participants initial baseline score and the post-study test score.

Table 1 illustrates that participants studying using the developed AI solution displayed a higher mean learning gain at $M = 59.45$ and $SD = 22.86$ compared to the participants studying using traditional textbook methods at $M = 25.84$ and $SD = 16.81$. The AI study group obtained a higher median value of 58.34 compared to the textbook study group's median value of 26.67.

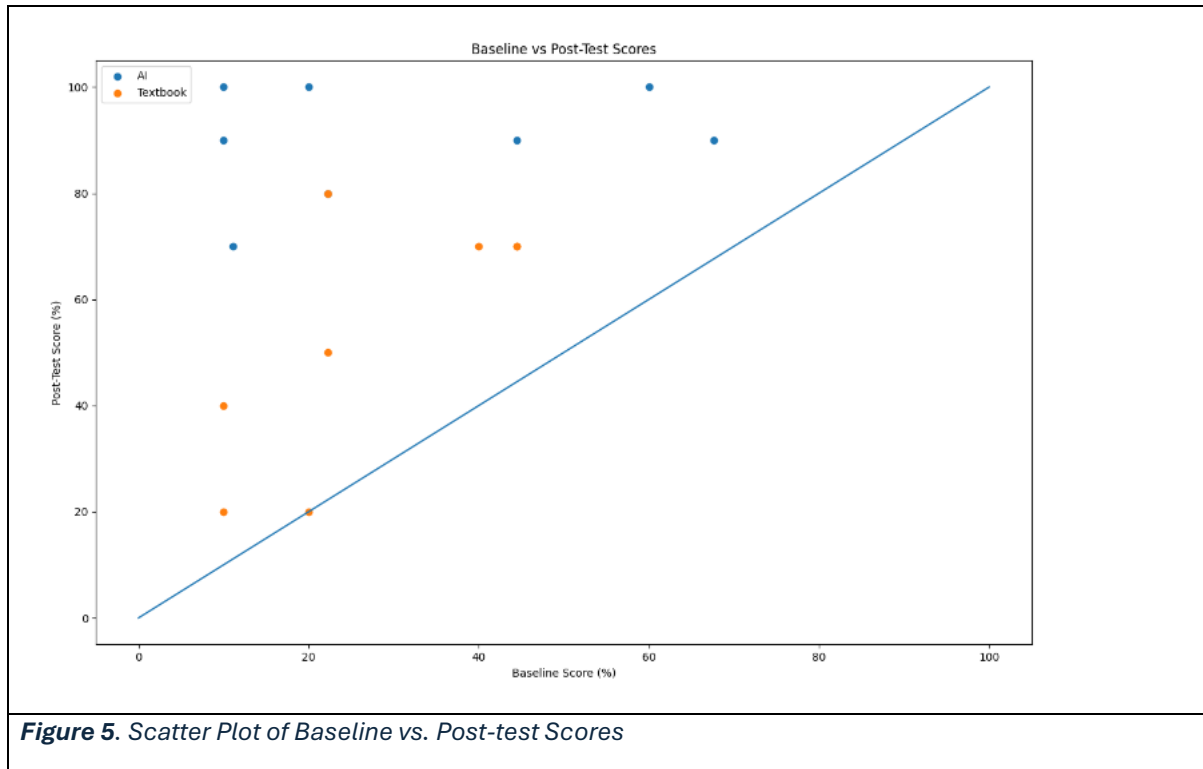
Table 1. Descriptive Statistics for AI-assisted study and Textbook Study

Mode of Study	Mean (percentage point increase) (Learning gains figure)	Standard Deviation (SD)	Median	Range
AI Study	59.45	22.86	58.34	23.33-90
Textbook	25.84	16.81	26.67	0-57.78

Differences between the group's performances were further reflected by the range of scores achieved. The AI study group's scores ranged from 23.33 to 90 percentage points increase compared to the textbooks study group's scores ranging from 0 to 57.78. 100% of the participants within the AI study group obtained positive learning gains as opposed to some participants within the textbook group showing minimal or no learning gains overall.

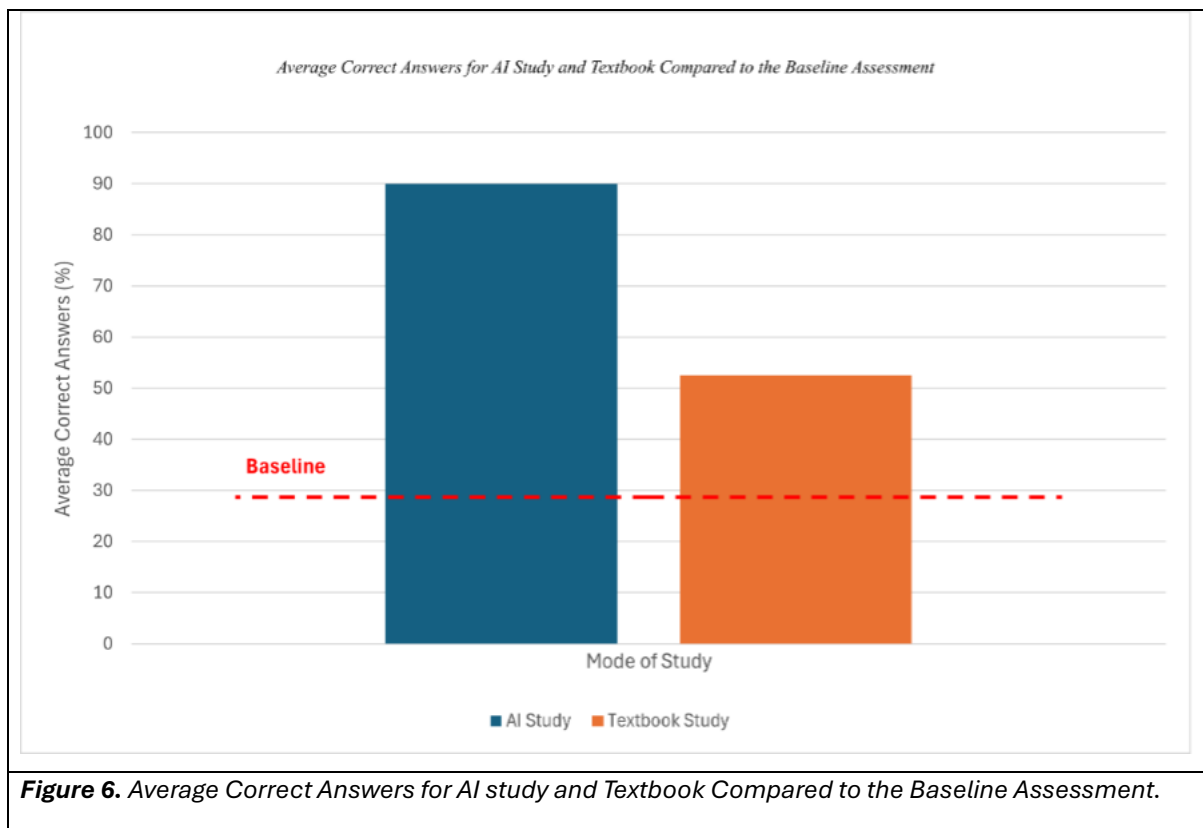
The higher standard deviation within the AI study group ($SD = 22.86$) reflected a higher variability in the achieved scores compared to those achieved in the textbook group ($SD = 16.81$).

Figure 5 demonstrates that the AI group, on average, achieved greater scores than the textbook group, with one participant in the textbook group showing no learning gain over the baseline.



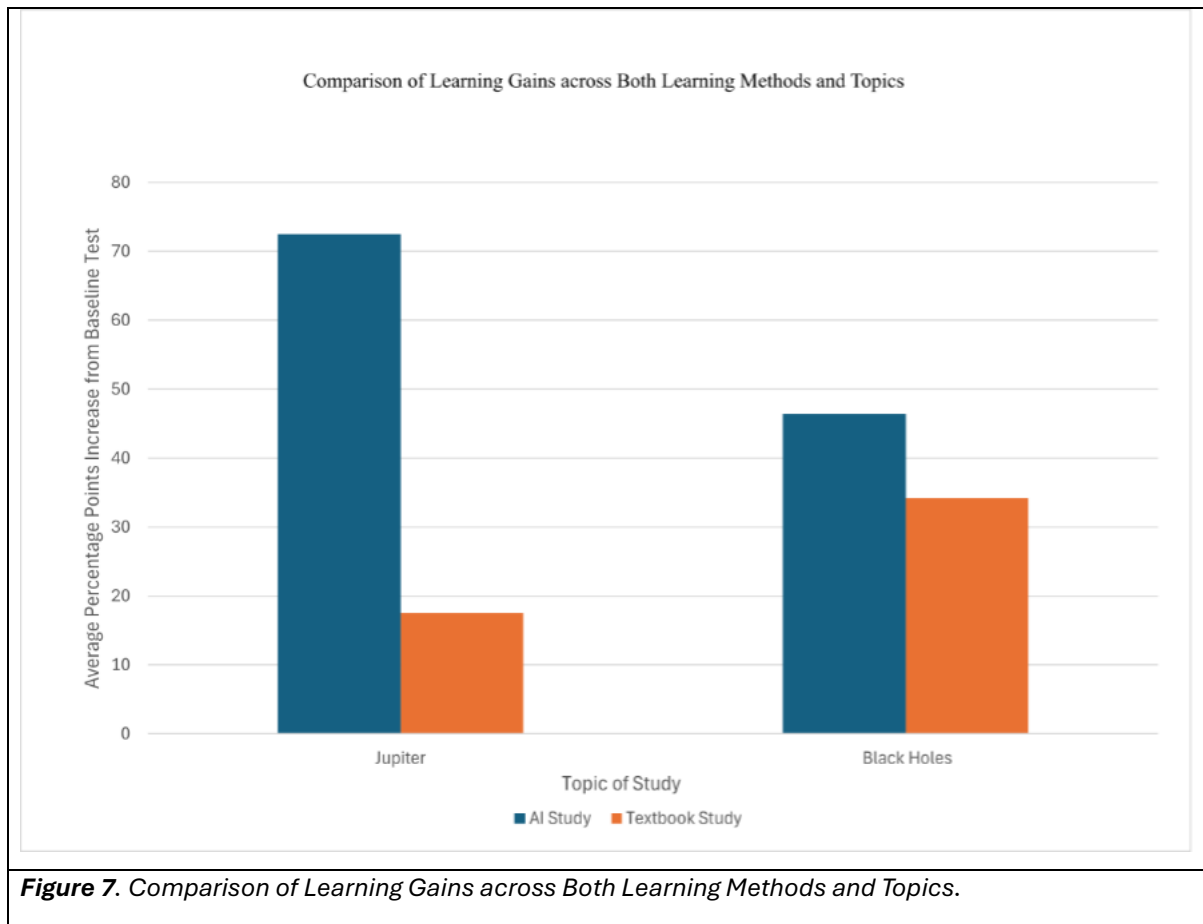
3.7.1.2 Baseline and Overall Performance

Figure 6 shows the average percentage of correct answers achieved by both study methods compared to the baseline assessment's average result (28.61%). The AI study group obtained a higher percentage of correct answers at 90% compared to the textbooks study group at 52%. Learning outcomes improved for participants using both learning methods. However, the AI study group accomplished higher learning outcomes overall, with 38 percentage points higher than the textbook group.



3.7.1.3 Topic Specific Results (Jupiter and Black Holes)

Figure 7 highlights the average percentage points increase (learning gains) for the AI and textbook study groups across both topics used within the study (Jupiter and Black Holes). As highlighted, when participants studied Jupiter, the AI study group performed better than the traditional textbook group at 72.50 percentage points increase from the baseline compared to a 17.50 increase. However, when participants studied black holes, the learning outcomes between methodologies were considerably closer. The AI study group's learning gains were calculated at 46.39 percentage points increase from baseline compared to the textbooks study group's learning gains being calculated at 34.17. Overall, the results show that when participants studied the topic "Jupiter", the learning outcomes between the learning methodologies had a higher percentage points difference compared to when they studied the topic "Black Holes".



3.7.1.4 Inferential Statistics

An independent samples t-test showed a statistically significant difference between learning methods with the p-value ($p=0.0048$). Using Cohen's d, a large effect size was identified at ($d=1.68$).

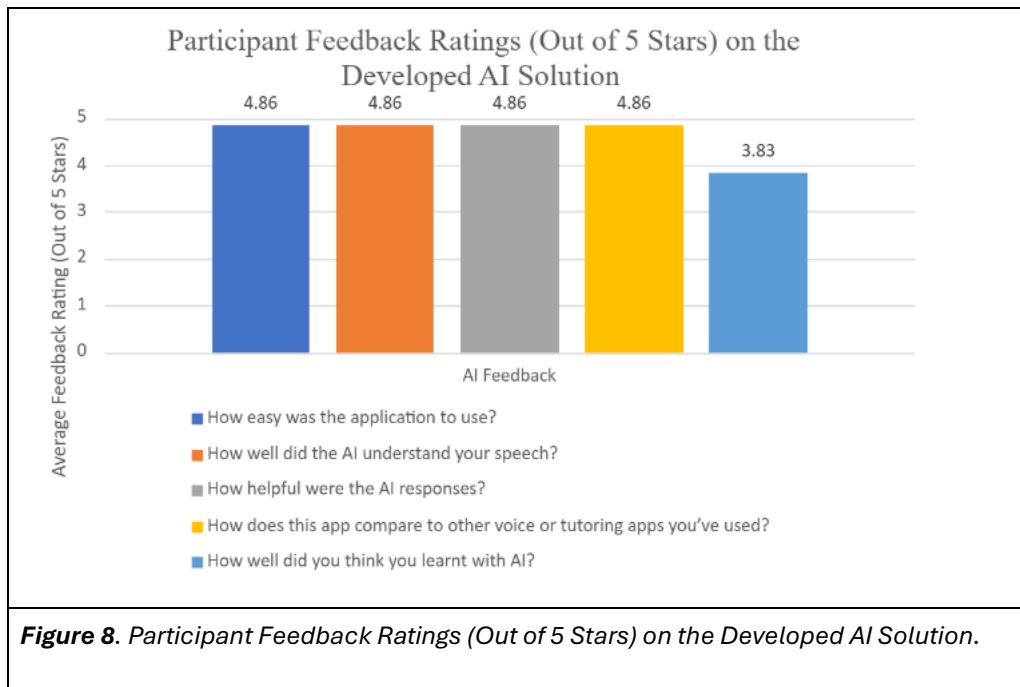
3.7.2 Qualitative Results

3.7.2.1 Feedback

Optional feedback form results were gathered from 6 participants, which were required to rate both learning methodologies on a scale of 1 to 5.

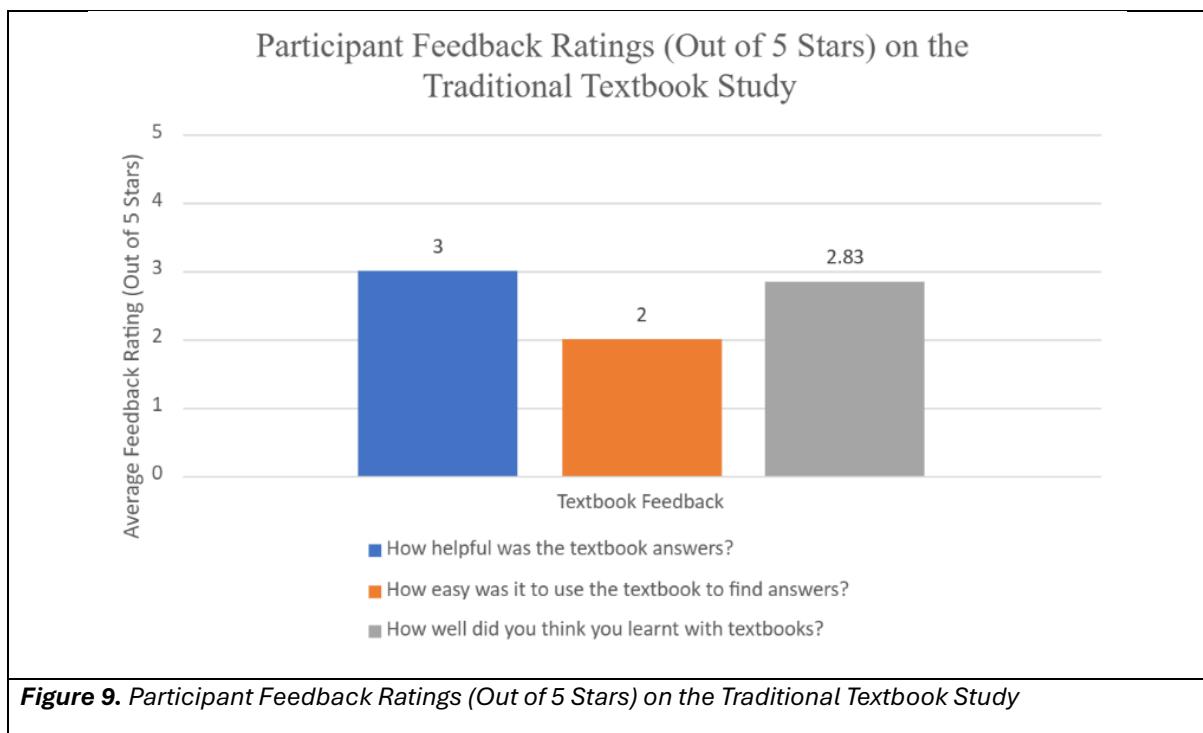
3.7.2.2 AI Feedback

Figure 8 illustrates participant's feedback regarding the developed AI solution (1-5). As shown, participants rated the AI application on average a 4.86 out of 5 for ease of use, how well the AI understood the user's speech, how helpful the AI responses were and how the application compares to other voice tutoring applications used previously. However, participants rated how well they thought they had learnt with the AI application a 3.83 out of 5.



3.7.2.3 Textbook Feedback

Furthermore, *Figure 9* highlights participant's feedback of the traditional textbook method (1-5). As demonstrated, participants rated the textbooks material on how helpful it was at providing support to formulate answers (M=3). Additionally, the textbooks ease of use was rated at 2 out of 5. Participants further rated how well they thought they had learnt with the material an average of 2.83 out of 5.



3.7.2.4 Thematic Responses

Participants were also required to answer open-ended questions which they would answer with opinion-based feedback about the learning methodologies used.

Two key themes were formed from the results gathered: 1) accessibility & knowledge retention 2) interactivity.

3.7.2.5 Accessibility and Knowledge Retention

Accessibility was a main recurring theme throughout the feedback gathered. 71% of participants reported that the developed AI solution was either easy to use, provided quick responses or gave accurate and clear answers. One participant stated that “AI is essentially more accessible” when compared to traditional textbook methodologies.

In contrast, 71% of participants outlined that it was difficult to gather answers whilst using the textbook methodology, with notable criticisms highlighting that there was “too much unnecessary information” and that it was “difficult to find answers”.

Several participants suggested that whilst using the AI application the need for active engagement was reduced and that knowledge retention was negatively affected. One participant noted that “the AI did not require me to think about the answers. I just copied what was said to me and put it out of my head straight away”.

In comparison, many participants highlighted that the textbook methodology required significantly higher cognitive engagement and reported that it resulted in increased knowledge retention. Participants noted that the textbook methodology “did not require as much revision as I retained more information within the 10 minutes” and “the words I have read in the textbook will stick in my mind for some time”.

3.7.2.6 Interactivity

Interactivity was a key theme identified, with participants reporting that the textbook methodology was “harder to navigate” than the AI system. Other participants noted that there was “too much unnecessary information” and that “even with the keyword finder, it was still difficult to find answers” when using the textbook material.

3.7.2.7 Participant Preference

Overall, the responses indicate that approximately 57% of the participants expressed preference for the AI application over the textbook alternative.

Raw participant datasets: Appendix J

4. Discussion

4.1 Overview

This study aimed to investigate the effectiveness of a developed AI astronomy application compared to traditional textbook methodologies in improving participant's learning outcomes. The results indicate that although the AI application resulted in greater learning gains within this study's limited timeframe, participants consistently reported lower perceived learning performance. This suggests that there was a discrepancy between the system's effectiveness and the participants' attitudes towards how they learned. Furthermore, this study partially met its research question, as the findings demonstrate that the AI application may be limited to only providing short-term learning outcomes, rather than providing an environment for sustained knowledge retention.

Overall, the findings support the existing literature that demonstrates that ITS-based systems can provide positive learning outcomes for participants (VanLehn , 2011; Ma et al., 2014). However, qualitative research provided deeper insights, suggesting that while the developed AI system improved participant learning performance, experience and perceptions of the system were not always positive.

4.2 Quantitative Findings and Statistical Analysis

4.2.1 Learning Gains

Participants in the AI group achieved a higher mean learning gain compared to the textbook group within the study's time constraints. This result suggests that the AI system was more efficient at providing information within this study's limited time for most participants. This is further reinforced by the study's median values, which reflect that learning outcomes within the AI group were consistent across all participants and were not influenced by a small percentage of outliers. Furthermore, the lower range of scores in the textbook group suggests that a small proportion of participants displayed minimal or no learning gains overall, whereas all participants in the AI group achieved positive learning outcomes.

In contrast, the AI group's higher standard deviation displayed a higher variability in learning gains compared to the textbook group. This finding suggests that, while the AI group showed consistently positive learning outcomes, the extent of the AI system's effectiveness was varied between participants. One explanation for this could be that increased learning gains from the AI system depend on a variety of participant characteristics, such as prior

astronomy knowledge, engagement with the system, or external influences (Schneider and Simonsmeier, 2025). As a result, there may be a limited generalisability to a wider population, as the AI system may not provide consistent learning outcomes across all instructional contexts.

4.2.2 Statistical Significance

An independent samples t-test indicated that there was a strong statistical difference between the AI group and the textbook group ($p=0.0048$). Therefore, this provided sufficient evidence to reject this study's null hypothesis (H_0).

Furthermore, the effect size was calculated to reveal the magnitude of the difference between the two learning methods. VanLehn (2011) conducted a meta-analysis of secondary research and reported a sizeable effect size of $d=0.76$ for intelligent tutoring systems. In contrast, this study conducted primary research and identified an effect size of more than double at $d=1.68$, indicating a very large practical effect size. Despite this, a direct comparison may not be applicable due to the difference in research methodology. Furthermore, the results of this statistical analysis need to be interpreted with caution as the small sample size ($n=8$) may inflate the results and limit the generalisability compared to VanLehn's (2011) broader research.

4.2.3 Factors Contributing Towards the AI System's Effectiveness

One factor contributing towards the effectiveness of the developed AI system is the increased accessibility compared to traditional textbooks in this investigation's conditions.

Observations during the investigation suggested that the AI system provided information efficiently and that this significantly reduced the time to search for answers. Consequently, this provided a key advantage for the AI condition in the limited timeframe, as participants could relay information from the system to the assessment. This suggests that similar systems may be effective as an assessment performance tool rather than supporting deep knowledge acquisition.

Another explanation for the effectiveness of the AI system within this study's time limitations is that it can generate responses that are tailored towards participants' prompts. Existing literature suggests that learning methodologies that directly address student misconceptions can improve conceptual understanding (Zeilik et al., 1997; Diakidoy and Kendeou, 2001). This study reinforces this finding, as the system's implemented guardrails can provide directed learning that aims to address learners' misconceptions. This suggests that the AI

system may be effective for learners who need additional support when trying to conceptualise this study's astronomy subject. However, this raises the question of whether the system supports active engagement with the material or if it is just providing learners with the answers.

4.3 The Contradiction Between Findings

This study shows strong quantitative empirical results supporting the effectiveness of the developed AI application within the study's constraints. However, through qualitative data collection, the results reveal a deeper contradiction between the system's actual performance and participants' perceptions. The participants reported positive ratings for the AI application in categories such as ease of use, system understanding and overall user experience. However, when participants rated the system in terms of perceived learning, there was a significant reduction in the score. This highlights an inconsistency between participant perceptions and the system's effectiveness that isn't revealed through the quantitative data alone.

As previously discussed, the developed AI application tailors responses directly towards learners' prompts. Although this presents advantages within this study's test constraints, learners may perceive that they have learned less due to the reduction in cognitive effort required to gather information. This could be an explanation for why the perceived learning ratings were lower, as instructional materials that provide immediate outcomes may be limited in supporting a greater understanding of content (Bjork and Bjork, 2011). This suggests that the application may be effective in providing accessible information for participants in the study's assessments; however, the system falls short when it comes to supporting deeper engagement with the material.

Observations made during the study period reinforce this perspective, as it was viewed that participants prioritised gathering information to complete the assessment's questions over using the system's interactive features. This was significant as it reduced the engagement participants had with the system, which may have resulted in reduced levels of conceptual understanding. Therefore, an explanation for why participants' learning perceptions were lower could be that assimilating "some but not all elements" of the subject material often limits the learner's ability to effectively understand concepts (Sweller, 1994). In this case, participants did not fully interact with the system's visualisations, resulting in a reduced depth of spatial understanding.

Participants also reported that they had copied answers from the application rather than interacting with the complete content. One participant stated, “I just copied what was said and put it out of my head straight away”, demonstrating the minimal effort that went into interacting with the system. This further indicates an implication in a wider educational context, as similar systems may provide effective support for efficient task assistance, but may be constrained in promoting cognitive engagement, which is suggested to potentially limit deeper understanding. Overall, this provides reasoning for why there was a discrepancy between findings, as participants may perceive that lower engagement correlates with reduced learning.

4.4 Topic-Specific Effectiveness

Furthermore, the effectiveness of the AI application varies between the astronomy topics within this study. One explanation for this could be that black holes tend to be more abstract in nature. They cannot be directly observed as they don’t emit light, and the only way to observe them is through the external orbiting matter (NASA, 2024). As a result, learners may find it difficult to develop a conceptual understanding of black holes within the system, as it may require spatial understanding to fully grasp the system’s visualisations (Cole et al., 2018). This raises a concern as similar systems may not provide sufficient representations that support learners’ spatial reasoning needs.

4.5 Evaluation of the Methodology and Findings

4.5.1 Strengths of this Study

A key strength of this study is its use of a Randomised Controlled Trial (RCT) design, which reduced bias by enabling participant characteristics to be randomly distributed between the initial conditions. Additionally, the design improved the study’s internal validity through providing a robust method for investigating the “cause-effect relationships” between the AI and Textbook conditions (Hariton and Locascio, 2018).

The crossover design further strengthened the study, as the participants experienced both learning methods, which reduced the influence that individual differences would have on the findings. It was particularly significant for identifying an effect size with a small participant pool, as the number of data points taken from the sample was effectively doubled (Capili and Anastasi, 2024).

4.5.2 Limitations of this Study

However, these strengths must be interpreted with caution, as the small sample size may have limited the reliability and reproducibility of the results. Although the RCT design randomly assigned participants into their initial conditions, there is still a possibility that participant characteristics such as prior knowledge and technological proficiency could have disproportionately influenced the measured outcomes. As a result, the effect size of this study may have been inflated, which overall limits the generalisability of these findings in an educational context. Additionally, the sample size only represented university students between the ages of 20 and 23, which limited the reliability of the findings, as the sample was not representative of a wider population.

The most significant limitation of this study was that no longitudinal learning outcome data were collected. As a result, the measured learning outcomes only reflect a short 10-minute study period. It's widely accepted that repeated learning spaced over time can result in enhanced learning outcomes (Dempster, 1989). However, this study did not implement repeated learning evaluations, limiting the findings to only short-term outcomes. Additionally, this study's assessment time limitations may not accurately reflect real-world educational environments, which limits the validity of the findings to this study's constraints.

4.5.3 Reliability, Reproducibility and Generalisability of the Findings

Overall, the results partly meet the research aim of this study, as the evaluated learning outcomes represent short-term performance rather than long-term learning. This supports existing literature that suggests that learning performance may not accurately reflect long-term understanding (Soderstrom and Bjork, 2015). As a result, the generalisability of these findings may not be applicable to other real-world educational contexts, as learning within these environments often occurs over an extended period. Although this study used a strong RCT design, the confidence in this study's findings is moderate, as the small sample size and short study duration limit the reliability and generalisability. Lastly, the results may not be reproducible in other learning environments, as the controlled test conditions and differences within individual participants may influence the findings.

4.6 Technical Evaluation

The participant feedback and testing suggested that the system was robust during the investigation phase. The initial testing produced sufficient performance with 90% of

functional tests passing successfully. Within the investigation phase, 100% of participants reported that no technical issues were encountered. One participant stated that the system provided “very accurate and fast” responses. This suggested that the system’s performance was consistent and reliable throughout both phases. The conducted usability tests identified that the audio would sometimes overlap, which overall reduced clarity in the responses. This issue was promptly resolved through iterative development, which targeted improving output precision. Functional tests discovered that background noise affected the accuracy of the responses generated, with the system occasionally pausing or producing incorrect results, which aligned with the design challenges highlighted in Pyae and Joelsson (2018). However, this issue could not be resolved as the API-based design approach limited development control over improving the system’s response accuracy. Overall, this demonstrated that although the system met its requirements of providing fast and accurate responses, the system could not produce consistent performance for every learning environment, suggesting a potential low feasibility for real-world deployment.

Participant feedback and observations demonstrated that the system had excellent usability, as it provided an intuitive, user-friendly UI. Many participants reported that the system was easy to use and accessible, indicating that the minimal design promoted simple navigation. Observations highlighted that the system was straightforward to learn, with users being able to interact with the different system elements with minimal instructions given. This suggested that the system met the ease-of-use and engagement requirements needed for responsive user interaction. In contrast, it was noted that several participants were copying answers from the system, highlighting that they were passively interacting rather than actively engaging with the content. This presented a key implication, as the easier-to-use the system was could have negatively affected participants’ learning.

The system was found to be effective as an assistive tool for astronomy study. However, it was limited by several key constraints. As discussed, the unresolved background noise limitation resulted in reduced input accuracy, affecting the reliability of the responses. Although this could not be resolved within this project's scope, future API updates may improve response performance. The quantitative topic-specific results revealed that the system was limited in displaying abstract astronomy visualisations, as it was found that the results varied between topics. This highlights a key implication of the application, as it may not be effective across all topics of astronomy study. Overall, as the system may not provide

guaranteed accuracy, it is limited in being able to be deployed in a real-world instructional setting.

4.7 Project Process Evaluation

As the project consisted of a development and research phase, it was essential to adhere to a project plan that ensured effective time management between both stages. Overall, the project predominately aligned with the initial project plan outlined in the TOR requirements (see Appendix B). However, some alterations had to be made during the process to keep the project streamlined. For example, there were some unexpected challenges encountered within the testing and development phases that needed resolving.

Although the project's TOR objectives such as the system development, participant testing and data analysis were all completed to the planned schedule, the testing and system refinements required additional time and resources, due to the usability tests revealing system issues. However, this provided valuable insights into iterative development processes.

Multiple technical and research skills were able to be developed during the process, including an improved understanding of React's component-based architecture and API integrations. The investigation phase provided deeper insights into experimental design decisions, data collection and critical evaluations.

Several challenges were encountered due to the project's constraints. The system's background noise limitation was out of the project's scope to resolve; however, it provided valuable lessons that not all technical limitations can be fully addressed. Additionally, the project's small sample size and study duration reduced the generalisability of the results gathered.

If the project process was repeated, more time would be allocated to planning an investigation with a larger participant pool. This would have resulted in greater reliability and generalisability in the findings. Furthermore, additional interactivity features would have been developed to enhance the learning outcomes and promote active engagement within the system.

Throughout the project, additional AI ethical risks had to be considered. Therefore, the project aligned with an implemented ethical considerations document, which ensured that relevant social, legal, ethical (SEL) challenges were addressed such as responsible use, transparency, accuracy and data privacy challenges.

5. Conclusion and Recommendations

This study aimed to evaluate the effectiveness of a developed voice-interactive AI learning platform compared with traditional e-textbook methodologies in increasing learning outcomes of participants in astronomy. The combination of qualitative and quantitative findings provided key perspectives on the learning outcomes and perceptions of participants' during the investigation phase. The findings of the study indicate that the AI system was more effective in providing short-term learning outcomes within the study's-controlled conditions (59.45 percentage points increase compared to 25.84). Despite this, participants reported lower learning perceptions, indicating that there was a discrepancy between the system's performance and participant attitudes. This indicates that the system was effective at retrieving information within a constrained timeframe, however, it was limited in providing participants with a greater understanding of the subject material, raising the concern of whether genuine learning occurred within the investigation. The conducted statistical analysis displayed a strong practical significance ($d=1.68$). However, these results should be interpreted with caution due to the small sample size ($n=8$) and short duration of this study. It can be concluded that this study partially met its research question, as the system's performance only reflected the study's limited conditions. Overall, this project contributes to wider astronomy research by highlighting the key trade-offs between accessibility and learner engagement that need to be considered when developing similar applications, as accessibility may negatively affect learning outcomes.

These findings highlight that future research should aim to collect longitudinal learning outcome data, which would further develop the understanding of how similar systems can influence learning over a sustained period. This research should focus on implementing intermittent assessments to evaluate how understanding develops over time, enabling a more in-depth analysis into the progression of learners' conceptual understanding.

Furthermore, future research should focus on improving the generalisability of these findings by increasing the sample size of the study, including recruiting a variation of ages, characteristics and demographics.

In regards of developing a system, upcoming work should prioritise developing an interface that would enhance the interactive functionalities, allowing for learners to gain a deeper immersion into the astronomy environment. Potential features could include task-based assessments and deeper explorative learning features.

Based on the statistical findings of this study, educational institutions should develop astronomy learning environments to assist students. However, these systems should supplement other learning methodologies rather than replace.

Overall, this study highlights the potential of voice-interactive AI-based learning environments and how they can support understanding in astronomy, while carefully considering the trade-offs that may arise when building these systems.

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7. Appendices

Practical Work One Drive Link: https://livenorthumbriaac-my.sharepoint.com/:f:/g/personal/w23012928_northumbria_ac_uk/IgAfvslh68M0QaW01MWNAAWEuAahdszqqSOY9iq-6k0tPdGA?e=jtQCDC

Appendix A:

Requirements Document: This highlights the non-functional and functional requirements that were essential for the research of the developed system.

Functional Requirements

FR1 - “The system should use voice-interaction as the main source of teaching output.”

FR2 - “The system should address student misconception by scaffolding learning, asking questions and providing feedback”

FR3 - “The system should have structured guardrails in place to ensure that accurate and consistent astronomy subject is being taught.”

FR4 - “The system should provide a three-dimensional astronomy environment to meet the spatial need of learners”

FR5 - “The system should allow users to ask astronomy-related questions”

FR6 - “The system should provide a chat history so users can view previously discussed topics”

FR7 - “The application should allow the user to navigate between astronomy planets”

FR8 - “The AI should be able to reply with context-driven answers based on previous chats”

Non-functional Requirements

NFR1 - “The system should aim to provide accurate, consistent and reliable responses to ensure a fair comparison during the investigation.”

NFR2 - “The system should be engaging and easy-to-use for all users.”

NFR3 - “The system should respond to user inputs within 2 seconds”

NFR4 - “The system should be robust and reliable for extended use”

NFR5 - “The system should be able to run on desktop computers”

NFR6 - “The system’s responses should be aligned with the astronomy subject”

Appendix B:

Terms of Reference (TOR): The following document was produced to define the project’s scope, goals and objectives.

Aim of Project:

The research aim of this project is to evaluate the effectiveness of a developed voice-interactive AI-based learning system compared to traditional e-textbooks methods in improving participants' learning outcomes when studying astronomy.

Objectives (max 10 objectives, max 150 words):

1. Conduct a 2000–2500-word literature review on Intelligent Tutoring Systems, astronomy misconceptions, interactive visual environments, and traditional e-textbook methodologies by 06/01/26
2. Create a requirements specification, including functional and non-functional requirements by 20/12/25
3. Create responsive Figma wireframes, through a completed, navigational prototype by 05/12/25
4. Implement a voice interactive OpenAI space prototype using React and JavaScript by 06/01/26
5. Complete relevant testing and refine the system by 16/01/26
6. Design a comparative study, with all relevant materials and procedures by 16/01/26
7. Complete at least 8 participant research assessments on my system by 27/02/26
8. Collect and organise participant results by 01/04/26
9. Analyse the collected data to compare learning outcomes between methods by 01/04/26
10. Evaluate the results and complete final 10,000-word project write-up by 07/05/26

Proposed Methodology / Work (max 300 words):

This project will be a combined process of software development and comparative research (with a primary focus on the research element). The technical component of this project focuses on how the AI-driven learning space application will be designed and implemented and the research side will then evaluate this software on the effectiveness of how it educates space subjects compared to e-textbooks.

For the technical component, a literature review will be conducted to understand the niche. A fully working voice-interactive AI system will be then designed by employing understanding of how to implement features using React and React Three fibre to create an interactive space environment and using OpenAI's API to integrate AI voice interactivity.

The research component will be a comparative study. Participants will initially take a baseline test to make sure they meet the requirements of the study and so that learning gains can be calculated. Participants will firstly use an astronomy application to revise for 10 minutes and then take an assessment. They will then be required to crossover and use an e-textbook to study for 10-minutes. Data received from both tests will then be analysed to compare differences in learning outcomes in both study methods. Evaluating how effective AI-driven learning can be in the context of space education.

Deliverables (max 50 words):

- Fully developed Interactive AI Space education application
- Prototypes
- Requirements specification

- AI ethical considerations document
- Testing Document
- Anonymised participant dataset
- Participant test materials (assessment/study task)
- Technical documentation (readme file)
- Final written project report

Assessment criteria (max 200 words):

Deliverable being assessed	Criteria	Evidence that will be provided
Interactive AI Space education application	A fully developed system enabling interaction with an AI tutor and navigation between planets in a 3D environment	System code, video of application working, readme file
Prototypes	Clear interactive user flow, including navigation, AI guidance, and task-based learning. High fidelity and low fidelity prototypes.	screenshots, annotations
AI ethical considerations document	Consideration of ethical AI usage (bias, reliability, transparency)	PDF document
Requirements specification	Relevant functional & non-functional requirements.	Requirements document
Testing document	Structured testing of system functionalities	Tables of test cases and results
Anonymised participant Dataset	Anonymised dataset of quantitative and qualitative participant data	Baseline and post-test results for AI and e-textbook conditions
Participant test materials	Relevant assessment materials aligned with research objective	Assessment questions and marking scheme
Technical Documentation and user manual	Clear instructions for system use	Readme file
Final written project report	Clear structure, in-depth analysis of results, and critical reflective evaluation of the research findings	Final 10,000 word report

Project Plan:

Semester 1

Week	Task	Dependencies
9	Begin Literature review; draft and gather academic resources	TOR goals and aims
10	Refine requirements and system design	Literature review insights
11	Complete Prototype; Start building React project setup and environment	Requirements, Prototype and Literature review research
12	Complete literature review; Implement 3D environment and AI voice interactivity/navigation features	TOR goals and aim; React project setup

Semester 2

Week	Task	Dependencies
1	Integrate all features (including chat history, etc.) and complete the program; test and refine the program; Develop documentation (readme) on the system	Main features implemented
2	Conduct testing on the program; log feedback; Start to develop the research assessment document	Working application with main features implemented
3	Fix major bugs; refine UI from feedback; Finish the research assessment document	testing document completed; Literature review and research
4	Start investigation with participants	Completed Research assessment document
5	Complete participant investigations	Finish the research assessment document; All participants have conducted their tests
6	Mark the assessments, Compile research findings, clean and anonymise the data	All participants have conducted their tests
7	Start comparative analysis; Start Final report	All research tests conducted and findings about the system have been discovered.
8	Continue data analysis	
9	Continue writing report	

10	Continue writing report	
11	Review and refine report	
12	Complete Comparative analysis and Final report	

Appendix C:

AI Ethical Considerations Document: This document was produced as the developed AI system had to be built with participants ethics in consideration. The following document is split into 6 different sub-sections (Responsible, Equitable, Traceable, Reliable, Governable, Trustworthy AI considerations)

Responsible AI Considerations

The use of AI required additional ethical considerations to ensure that the system was used responsibly by participants within the investigation phase. The system was developed with the following considerations in place:

Responsible AI:

Responsibility for the system's behaviour was maintained through the backend. Guardrails were implemented to limit the AI tutor to only respond with educational astronomy content. The system was designed to be used within an education context. The developer of the application was responsible for how the system was deployed, maintained and used within the study.

Equitable AI:

Bias was minimised through imposing strict guardrails onto the AI tutor, restricting its responses to astronomy-related content. This reduced the likelihood of inaccurate or bias responses. However, it was not guaranteed that all these inaccuracies were eliminated, so the system provided participants with a disclaimer highlighting that it may still produce incorrect results. Additionally, the participant information sheet outlined that "Participants are potentially exposed to wrong information proposed from the AI", which was given to participants prior to them starting the investigation.

Guardrails:

Only provide scientifically accurate information.
If unsure, say you do not know.
Do not guess or fabricate answers.

Traceable AI:

Within this project, the systems architecture, development process, API usage and external libraries were all documented throughout the report. Additional documentation such as, the appendix, README file and additional system documentation further supported traceability. This ensured transparency and reproducibility throughout the project.

Reliable AI:

The system was defined to only be used as an AI astronomy Tutor. Reliability testing within this report included functional, integration, edge case and usability tests to ensure that the systems effectiveness was reliable within this projects scope. If the system did not meet the reliability requirements, a higher accuracy model would have been selected. However, the selected model produced sufficient results within the testing phase, so was preferred.

Guardrails in place to help support reliable results:

You are Astro, a space tutor.


```
Only provide scientifically accurate information.  
If unsure, say you do not know.  
Do not guess or fabricate answers.
```

Governable AI:

Backend guardrails were implemented to restrict unintended use. Testing highlighted that the system was able to detect unintended consequences and revert the user back to normal use. The backend instruction used in this case was:

```
Be concise and conversational. Make sure you only answer questions about space, if it's not about space say that you don't know and move on.
```

Trustworthy AI

The system aimed to provide users with trustworthy responses:

```
Only provide scientifically accurate information.  
If unsure, say you do not know.  
Do not guess or fabricate answers.
```

The system did not store any personal data and the system's chat history from the investigation phase was not retained to stay in line with the Data Protection Act (2018).

The OpenAI API does not store data for training purposes.

Appendix D:

Participant Information Sheet & Consent Form: This document was in place to ensure that this study aligned with the ethics agreed.

The participant information sheet handed out to participants:

Consent Form:

PARTICIPANT CONSENT FORM

Project Title: **Comparative study on effectiveness of AI education in astronomy**

Principal Investigator (student): **elliott.massey-1**

Student Email: **w23012928@northumbria.ac.uk**

Please tick or initial where applicable –

- I have carefully read and understood the Participant Information Sheet. ☐
- I consent to video recordings being made of me during this study in line with the activities outlined in the information sheet. ☐
- I consent to audio recordings being made of me during this study in line with the activities outlined in the information sheet. ☐
- I have had an opportunity to ask questions and discuss this study and I have received satisfactory answers. ☐
- I understand I am free to withdraw from the study at any time, without having to give a reason for withdrawing, and without prejudice, however my data may be kept and/or stored in an anonymized form if analysis has already started, in line with the information provided in the information sheet. ☐
- I agree to take part in this study. ☐
- I also consent to the retention of this data under the condition that any subsequent use also be restricted to research projects that have gained ethical approval from Northumbria University. ☐
- I agree to the University of Northumbria at Newcastle recording and processing this information about me. I understand that this information will be used only for the purpose(s) set out in the information sheet supplied to me, and my consent is conditional upon the University complying with its duties and obligations under the Data Protection Act 2018 which incorporates General Data Protection Regulations (GDPR). You can find out more about how we use your information here [Privacy Notices](#) ☐

Name/Signature of participant.....

Date.....

Participant Information Sheet:

Comparative study on effectiveness of AI education in astronomy

Participant Information Sheet

You are being invited to take part in this research study. Before you decide it is important for you to read this leaflet so you understand why the study is being carried out and what it will involve.

Reading this leaflet, discussing it with others or asking any questions you might have will help you decide whether or not you would like to take part.

What is the Purpose of the Study

AI is a growing sector, that's widely used in study. The aim is to show whether it can be used as an educational tool for moving forwards.

Why have I been invited?

University peers (over 18, under 60)

Do I have to take part?

No. It is up to you whether you would like to take part in the study. I am giving you this information sheet to help you make that decision. If you do decide to take part, remember that you can stop being involved in the study whenever you choose, without telling me why. You are completely free to decide whether or not to take part, or to take part and then leave the study before completion.

Please note, that if you withdraw:

Participants can withdraw their data, but only before the analysis process has begun.

What will happen if I take part?

1. Participants will have to take a pre-test about Topic A & B. This establishes a baseline for study.
2. Participants will only be told the topic on the day, so that no one can pre-study.
3. Participants will then be separated into two groups:
 - One group will use AI on Topic A; the other group will use digital textbooks on Topic A
4. Participants will then take a post-test containing questions based on Topic A. Group 1's test scores will reflect AI learning, and Group 2's test scores will reflect digital textbook study.

PARTICIPANT INFORMATION SHEET



5. Groups will switch methods for Topic B

6. Participants will take a post-test containing questions based on Topic B. Now Group 1's score reflects textbook learning, and Group 2's score reflects AI learning.

What are the possible disadvantages of taking part?

1. Participants are potentially exposed to wrong information proposed from the AI (I will provide a clear user-facing disclaimer stating that some information may not always be correct or may have bias)
2. Participants may be exposed to bright lights and flashing images (I will provide an epilepsy warning when loading the webpage)
3. Participants may expose personal information to AI (I will make sure the AI doesn't retain data to train models, data is deleted for protection)
4. Participants data being exposed in transit (I will always use Https, OpenAI's SDK does this by default)

What are the possible benefits of taking part?

1. Participants may have an enhanced learning experience
2. Participants may have improved knowledge retention and understanding of space topics
3. Participants can contribute to real research in a developing sector
4. Participants can contribute to broader societal impact
5. Participant skill development

Will my taking part in this study be kept confidential and anonymous?

Yes. The information you provide will be cleaned in the final dataset so that only an ID number shows, not your name.

If you are being recorded (audio or video), your name will not be written on the recorded interviews, or on the typed-up versions of your discussions from the interview, and your name will not appear in any reports or documents resulting from this study.

The consent form you have signed will be stored separately from your other data. The data collected from you in this study will be confidential. The only exception to this confidentiality is if the researcher feels that you or others may be harmed if information is not shared.

How will my data be stored, and how long will it be stored for?

All paper data including the questionnaires, the typed-up, transcripts from your interview, and your consent forms will be kept in locked storage. All electronic data including any recordings from interviews, will be stored on the University OneDrive, which is password protected. All data will be stored in accordance with Northumbria University guidelines and the Data Protection Act (2018). Only the student researcher, their supervisor and the marking team will have access to this data. Data will be stored for a maximum of 3 years and then removed.

What categories of personal data will be collected and processed in this study?

PARTICIPANT INFORMATION SHEET



Participants' first names or participant University ID codes, age (for demographic purposes), and optionally an email address for communication or withdrawal requests. No other personally identifiable data (e.g., full name, address, ethnicity) will be collected. 1. Pre-test and post-test scores on learning topics. 2. Responses to post-learning surveys (engagement, perceived understanding, usability) 3. Interaction logs from the AI tutor system (e.g., time spent, number of interactions, navigation data) 4. General demographic information such as age group or prior familiarity with astronomy topics

What is the legal basis for processing personal data?

Processing of this data is necessary for scientific research purposes.

Who are the recipients or categories of recipients of personal data, if any?

The data collected in this study will be processed by the named student researcher, and will be available to their project supervisor and the module tutor.

What will happen to the results of the study and could personal data collected be used in future research?

In rare cases, the general findings might be reported in a scientific journal or presented at a research conference, however the data will be anonymized and you or the data you have provided will not be personally identifiable, unless we have asked for your specific consent for this beforehand.

The findings may also be shared with other organizations/institutions that have been involved with the study. We can provide you with a summary of the findings from the study if you email the researcher at the address listed below.

Who is Organizing and Funding the Study?

This project is organised by Northumbria University, within the KV6013 Undergraduate Individual Project module.

Who has reviewed this study?

The research project, has been approved under the Module Level ethics approval sought for KV6013 in Northumbria University's Ethics Online system. It has been reviewed in order to safeguard your interests, and have granted approval to conduct this study.

What are my rights as a participant in this study?

A right of access to a copy of the information comprised in their personal data (to do so individuals should submit a Subject Access Request); a right in certain circumstances to have inaccurate personal data rectified; and a right to object to decisions being taken by automated means.

If you are dissatisfied with the University's processing of personal data, you have the right to complain to the Information Commissioner's Office. For more information see [the ICO website](#).

PARTICIPANT INFORMATION SHEET



Contact for further information:

This project is being conducted by *elliott.massey-1* within the context of a student project in module KV6013.

They can be contacted at:
w23012928@northumbria.ac.uk

For other queries, contact the module tutor (ethics approval holder for the module):

tom.bartindale@northumbria.ac.uk

Name and contact details of the Records and Information Officer at Northumbria University: *dp.officer@northumbria.ac.uk*

*You can find out more about how we use your information at:
www.northumbria.ac.uk/about-us/leadership-governance/vice-chancellors-office/legal-services-team/gdpr/gdpr-privacy-notice/
or by contacting a member of the research team*

Appendix E:

Assessment Questions (example of each type): This shows an example of each type of assessment question that participants encountered:

Multiple-choice questions with one correct answer:

7. How *long* is *one day* on Jupiter in hours?

- ☐ 9.9 hours
- ☐ 16.9 hours
- ☐ 24 hours

True/False questions:

6. Does our Milky Way, have a black hole at its *center*? True or False? ...

- ☐ True
- ☐ False
- ☐ I don't know

Open-ended questions:

7. What is the *disk of gas and dust* surrounding a black hole called?

Enter your answer

Full Assessment Questionnaires:

Baseline:

Section 1 - The Planet Jupiter

- In this section of the questionnaire you will be asked **10 questions** based on the planet **Jupiter**.
- Please answer all questions using your **own knowledge** rather than searching for resources as this will keep the results of this study valid.
- If at any time you are stuck on a question or want to move onto the next question please type into the question box either of the following: "I don't know", "N/A" or just leave the question box blank.
- You will **NOT** be given any results from the questionnaire.

Please read through each question carefully before answering the question.

6. Why does Jupiter have *rings* that are in orbit?

Enter your answer

7. Jupiter is named after a Roman God. Please tell me *who was* Jupiter, and *why was* this name *appropriate* for the planet?

Enter your answer

8. How many planets *away from the Sun* is Jupiter?

Enter your answer

9. What is the *radius* of Jupiter in km?

Please tick the box of the answer you think it is

- ☐ 69,911 km
- ☐ 52,200 km
- ☐ 27,327 km

10. Do you believe that Jupiter can *hold life*? why or why not?

Answer in the following format: **(I do/ I don't)** believe that Jupiter can hold life because...

Enter your answer

11. **How long in days does it take Jupiter to orbit the Sun once?**

How long is one year on Jupiter?

Enter your answer

12. **What is the average temperature(°C) of Jupiter's upper atmosphere?**

Enter your answer

13. **Why is Jupiter's gravity stronger than Earth's?**

Enter your answer

14. **What makes Jupiter's storms stronger than Earth's?**

Enter your answer

15. **What is the name of Jupiter's largest moon?**

Enter your answer

16. Are you **confident** in the answers you provided in **section 1**? *

☐ Yes

☐ No

Section 2 - Black Holes

- In this section of the questionnaire you will be asked **10 questions** based on **black holes**.
- Please answer all questions using your **own knowledge** rather than searching for resources as this will keep the results of this study valid.
- If at any time you are stuck on a question or want to move onto the next question please type into the question box either of the following: "I don't know", "N/A" or just leave the question box blank.
- You will **NOT** be given any results from the questionnaire.

Please read through each question carefully before answering the question.

17. What *important discovery* related to black holes did scientists first detect in 2015?

Enter your answer

18. How *close* is the *closest black hole* to Earth in the measurement light-years?

- ☐ 3,000 light-years away
- ☐ 10,000 light-years away
- ☐ 100,000 light-years away

19. What does the term *spaghettification* in the context of black holes mean?

Enter your answer

20. Can *black holes emit large amount of harmful radiation*? True or False?

- ☐ True
- ☐ False
- ☐ I don't know

21. What *happens to time* as you get closer to a black hole?

- ☐ Time appears to pass normally for you, but distant observers see your time slow down.
- ☐ Time appears to pass faster for you, and distant observers see your time speed up.
- ☐ Time dilation does not occur.

22. What is the name of the force that pulls objects into a black hole?

Enter your answer

23. What type of black hole forms from the collapse of a massive star?

Enter your answer

24. What part of a black hole forms marks the point of no return?

Enter your answer

25. What do scientists call the region around a black hole where gravity is very strong, but there is still possibility of escape?

Enter your answer

26. Why are black holes particularly difficult to directly observe?

Enter your answer

27. Are you **confident** in the answers you provided in **section 2**?

- ☐ Yes
- ☐ No

Jupiter Post-study Questionnaire:

3. How many recognized *moons* does Jupiter have?

"A moon is an object that orbits a planet or something else that is not a star." (National Geographic)

Enter your answer

4. Can you name at least *two* of Jupiter's *moons*? please provide their names:

Enter your answer

5. What is *the composition of Jupiter*?

What are the main materials or substances that make up the planet?

Enter your answer

6. What is Jupiter's *giant red spot*?

Enter your answer

7. How *long* is *one day* on Jupiter in hours?

☐ 9.9 hours

☐ 16.9 hours

☐ 24 hours

8. What type of planet is Jupiter?

⋮

- ☐ Rocky planet
- ☐ Gas Giant
- ☐ Ice Giant

9. What causes Jupiter's light and dark cloud bands?

Enter your answer

10. What is the main gas found in Jupiter's atmosphere?

Enter your answer

11. Name one spacecraft that has studied Jupiter up close?

Enter your answer

12. Give a reason for why Jupiter is sometimes called a "failed star"?

Enter your answer

13. Are you **confident** in the answers you provided in the **Jupiter questionnaire**? *

- ☐ Yes
- ☐ No

Black holes post-study questionnaire:

3. What is a black hole?

Enter your answer

4. What is the *boundary* around a black hole called?

Enter your answer

5. Name *one type* of black hole?

Enter your answer

6. Does our Milky Way, have a black hole at its *center*? True or False?

- ☐ True
- ☐ False
- ☐ I don't know

7. What is the *disk of gas and dust* surrounding a black hole called?

Enter your answer

8. What is one danger of getting too close to a black hole?

Enter your answer

...

9. What tool did scientists use to take the first picture of a black hole in the year 2019?

Enter your answer

10. Why cannot light escape from a black hole?

Enter your answer

11. What happens to an object when it crosses the event horizon of a black hole?

Enter your answer

12. What is the name given to the center point of a black hole where density becomes very high?

Enter your answer

13. Are you **confident** in the answers you provided in **section 2**?

☐ Yes

☐ No

Appendix F:

Assessment marking scheme example:

Marking Scheme

Black Holes

Question 1: **What is a black hole?**

ANSWER: A black hole is a region in space, where gravity is so strong that not even light can escape. (NASA STEM Team, 2014) (1 mark)

Question 2: **What is the *boundary* around a black hole called?**

ANSWER: Event horizon (Harvard, 2024) (1 mark)

Question 3: **Name *one type* of black hole?**

ANSWER: Any of the following: Stellar, Stellar mass, Supermassive, Intermediate, Primordial. (NASA, 2024) (1 mark)

Question 4: **Does our Milky Way, have a black hole at its *centre*? True or False?**

ANSWER: True (Wikipedia Contributors, 2018) (1 mark)

Question 5: **What is the *disk of gas and dust* surrounding a black hole called?**

ANSWER: Accretion disk (Wikipedia Contributors, "Accretion Disk", 2019)

Question 6: **What is one *danger* of getting too close to a black hole?**

ANSWER: Spaghettification (Gohd, 2023) (1 mark)

Question 7: **What tool did scientists use to take the first picture of a black hole in the year 2019?**

ANSWER: Event Horizon Telescope (NASA, "First Image of a Black Hole - NASA Science", 2019) (1 mark)

Question 8: **Why cannot light escape from a black hole?**

ANSWER: Spacetime curvature ("General Relativity - Why Can't Light Escape from a Classical Black Hole?")

Question 9: **What happens to an object when it crosses the event horizon of a black hole?**

ANSWER: Spaghettification, passing no point of return where gravity prevents escape, accelerating it toward the centre singularity. (Gohd, "What Happens When Something Gets 'Too Close' to a Black Hole? - NASA Science", 2023)

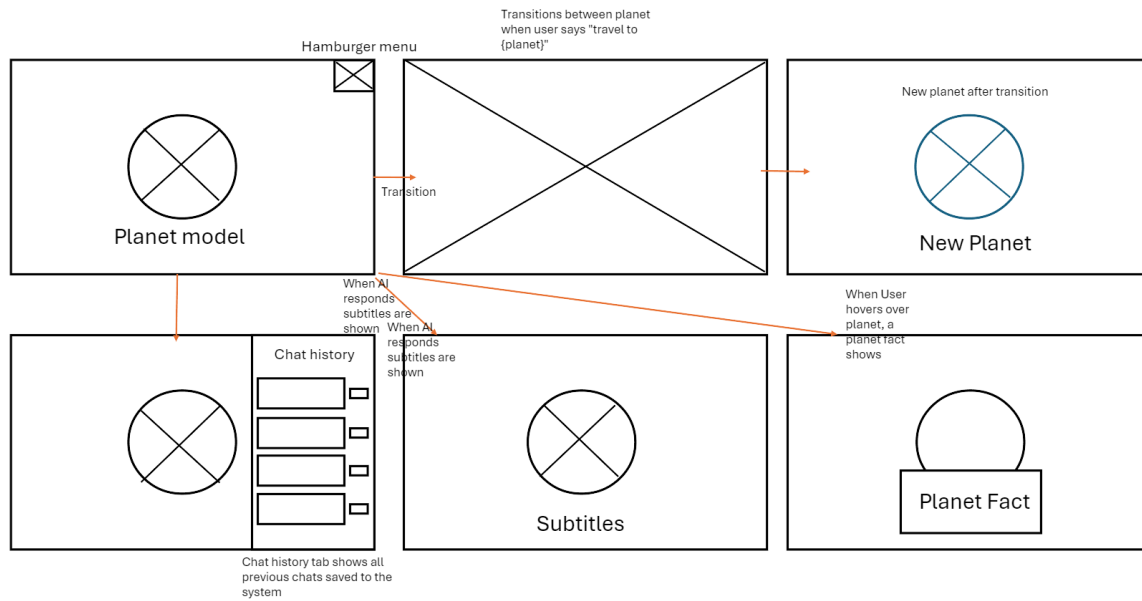
Question 10: **What is the name given to the centre point of a black hole where density becomes very high?**

ANSWER: Singularity ("Gravitational Singularity", 2020)

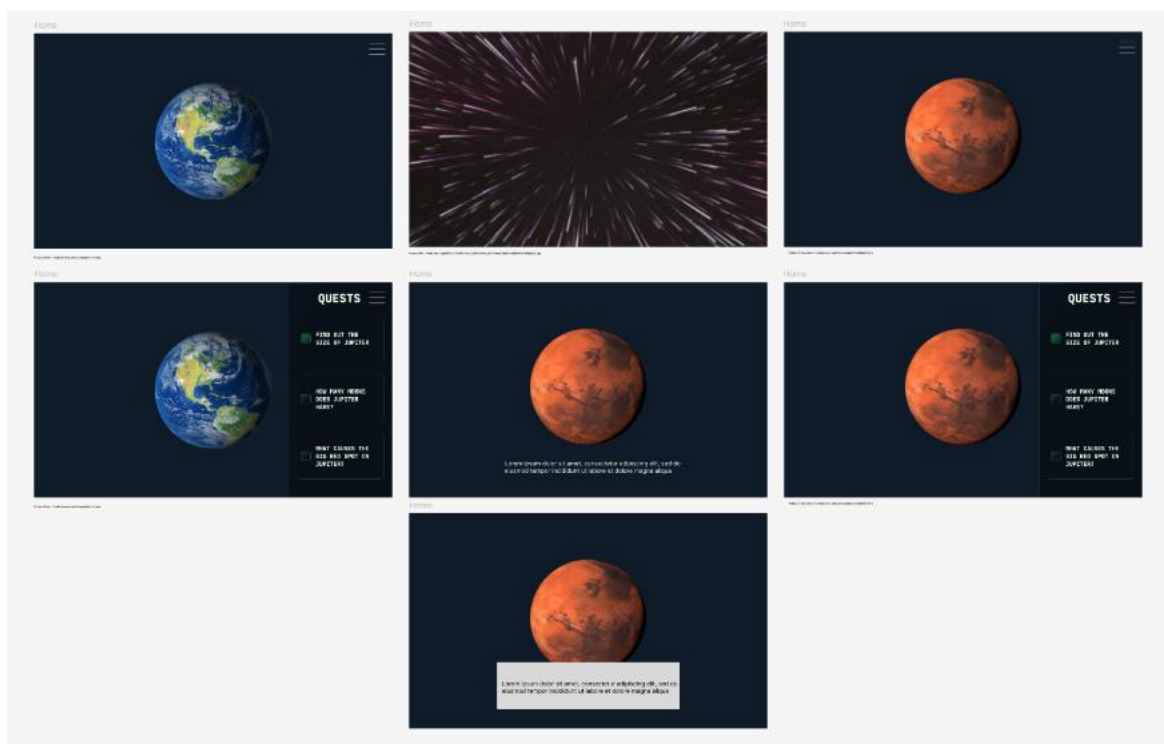
Appendix G:

System Prototypes and readme file:

Low Fidelity Prototype



High Fidelity Prototype:



Readme file:

Voice-Interactive AI Astronomy Learning Platform

Overview

Developed Voice-interactive AI astronomy learning platform.

This is the system that participants will use within the investigation phase of my project. It exists to provide learning support to participants and will have its effectiveness assessed against traditional textbook methodologies.

Video Tutorial

If you would prefer a guided walkthrough of the system, a video tutorial is available at:

OneDrive > Developed System > Video Presentation & Tutorials

This demonstrates how to set up, run, and use the application.

Features

1. Planet orbit and zoom functionality
2. Travel to any planet in the solar system by asking the AI to “travel to {planet}”
3. View previous chat history in the chat history drawer
4. Hover over the planet to see a fact about the current planet

Tech Stack

Frontend

- React
- React Three Fiber

Backend

- Node.js
- Express

Assets

- Planet textures by James Hastings-Trew
- Source: <https://planetpixeemporium.com/>
- These textures are free to use in personal and commercial projects but may not be redistributed as standalone assets without permission

Installation / Setup

Prerequisites

- A code IDE (preferably Visual Studio Code)
- Node.js (v18 or higher)
- `npm`
- A running backend server on `http://localhost:3001` (used to generate OpenAI session keys)
- OpenAI API key (configured in the backend)
- A modern browser (Chrome recommended)
- Microphone access enabled
- Stable internet connection

Steps

1. Install Visual Studio Code: <https://code.visualstudio.com/>
2. Install Node.js: <https://nodejs.org/en/download>
3. Download code files from:
OneDrive > Developed System > System Code Files
4. Unzip the code files
5. Open the folder in Visual Studio Code
6. Open a terminal in VS Code
7. Install dependencies:

`npm install`

8. Install additional packages:

`npm install react react-dom three @react-three/fiber @react-three/drei express`

9. Start backend (new terminal):

`cd server`

`node server.js`

10. Start frontend (new terminal):

`npm run dev`

11. Open the app:

- Hold **CTRL** and click:

`http://localhost:5173/`

Running the Application

12. Load application
13. Accept disclaimer
14. Press **connect** button
15. Communicate with the AI tutor
16. To travel to a planet, say:
 "travel to {planet}"
 (Only planets in the solar system are supported)

Usage

1. Open app in browser (<http://localhost:3000> or <http://localhost:5173>)
2. Accept the disclaimer
3. Connect to the AI assistant
4. Ask questions using your voice (e.g. *"tell me about Mars"*)
5. Planet transitions – AI mentions trigger travel
6. Interact with the 3D scene – orbit and zoom
7. View planet facts – hover over planet
8. Chat history – open drawer (top right)
9. Disconnect AI – press disconnect button

Project Structure

```
├─ dist/
├─ node_modules/
├─ public/
|   ├─ textures/
|   ├─ readme.md
|   └─ references
|   └─ vite.svg
|
├─ server/
|   └─ .env
|   └─ server.js
|
└─ src/
```

```
| ├── assets/
| ├── components/
| ├── App.jsx
| ├── main.jsx
| ├── index.css
| └── useRealtimeAI.js
|
└── package.json
└── vite.config.js
```

Configuration

API keys should be included in project files. However:

The API key (please keep this safe):

[REDACTED FOR SECURITY]

If it is not already in the .env file, copy and paste it into it.

Known Issues / Limitations

1. Background noise (use in a quiet room)
2. Multiple inputs (speak clearly)
3. Multiple tabs = multiple responses (use one tab only)
4. Planet transition may stop due to rate limits
→ Disconnect and reconnect to fix

Contributors

Elliot James Massey

References

Image Credits

Planet textures by James Hastings-Trew

<https://planetpixlemporium.com/>

OpenAI API

<https://platform.openai.com/>

React / Three.js

React — <https://react.dev/>

React Three Fiber — <https://docs.pmnd.rs/react-three-fiber/>

ChatGPT

This project was developed for the investigation phase.

Some code debugging and assistance was supported using ChatGPT (OpenAI).

Some planetary images were generated by ChatGPT (not used in final version but available in /textures).

<https://openai.com/>

Appendix H:

Key Code Implementations: These code screenshots are directly from the system and are explained in the implementation section of the dissertation:

connectAI:

```
// connects to the AI service using WebRTC and handles the audio/transcript stream
async function connectAI() {
  try {
    // fetch a temporary client key from the backend for authentication
    const response = await fetch("http://localhost:3001/session");
    const data = await response.json();
    const clientKey = data.client_secret?.value || data.client_secret;

    // create a peer-to-peer WebRTC connection to the AI service
    const pc = new RTCPeerConnection();
    pcRef.current = pc; // store in ref so connect/disconnect functions will work
    // create an audio element to play the AI's voice stream
    const audioEl = document.createElement("audio");
    audioEl.autoplay = true; // play as soon as the stream is received
    pc.ontrack = (e) => (audioEl.srcObject = e.streams[0]); // attach the AI audio stream to the audio element

    // data channel receives realtime transcript messages from the AI service....used for subtitles and planet detection
    const dc = pc.createDataChannel("oai-events");

    //event handler for incoming messages on the data channel from the AI service
    dc.onmessage = (event) => {
      try {
        const msg = JSON.parse(event.data);

        // transcript increment new text to the current response display
        if (msg.type === "response.audio_transcript.delta") {
          setResponseText((prev) => prev + msg.delta);
          currentTranscriptRef.current += msg.delta; // get full transcript
        }
      }
    }
  }
}
```

- The temporary key is fetched from the backend to establish a web RTC connection. This is created so the key is never exposed to the frontend.

RTCPeerConnection:

```
// create a peer-to-peer WebRTC connection to the AI service
const pc = new RTCPeerConnection();
pcRef.current = pc; // store in ref so connect/disconnect functions will work
// create an audio element to play the AI's voice stream
const audioEl = document.createElement("audio");
audioEl.autoplay = true; // play as soon as the stream is received
pc.ontrack = (e) => (audioEl.srcObject = e.streams[0]); // attach the AI audio stream to the audio element

// data channel receives realtime transcript messages from the AI service....used for subtitles and planet detection
const dc = pc.createDataChannel("oai-events");
```

- creating a peer-to-peer connection to the open AI API

- Creating an audio element
- Making sure it's played when the stream is received
- Attaching that audio element to the stream
- This ensures that the audio is played immediately when the stream comes through
- A data channel is created to handle transcripts in isolated away from the audio, this allows for the subtitles to update without affecting the audio.

dc.onmessage:

```

//event handler for incoming messages on the data channel from the AI service
dc.onmessage = (event) => {
  try {
    const msg = JSON.parse(event.data);

    // transcript increment new text to the current response display
    if (msg.type === "response.audio_transcript.delta") {
      setResponseText((prev) => prev + msg.delta);
      currentTranscriptRef.current += msg.delta; // get full transcript
    }

    // transcript done: save to history, check for planet mention
    if (msg.type === "response.audio_transcript.done") {
      const answer = currentTranscriptRef.current;
      const question = lastUserQuestionRef.current || userTranscriptRef.current;

      if (question || answer) addQA(question, answer);

      // check if a planet was mentioned and and trigger a transition
      const mentionedPlanet = detectPlanetMention(answer);
      if (
        mentionedPlanet &&
        mentionedPlanet !== currentPlanetRef.current
      ) {
        onPlanetDetected(mentionedPlanet);
      }

      // clear subtitle after a duration proportional to response length
      clearTimeout(window.subtitleTimeout);
      const duration = Math.max(4000, answer.length * 60);
      window.subtitleTimeout = setTimeout(() => setResponseText(""), duration);

      currentTranscriptRef.current = ""; // reset transcript for the next response
    }

    // get the user's spoken transcript
    if (msg.type === "input_audio_transcript.delta") {
      userTranscriptRef.current += msg.delta;
    }

    // when user transcript is complete, store it as the last question
    if (msg.type === "input_audio_transcript.done") {
      lastUserQuestionRef.current = userTranscriptRef.current;
      userTranscriptRef.current = "";
    }
  } catch {}
}

```

- Receiving full transcript of ai response
- Saving to history
- Checking for planet detection (goes to a planet detection function)

Planet detection function:

```

// detects if a planet is mentioned in the AI response text
function detectPlanetMention(text) {
  const s = (text || "").toLowerCase(); // convert to lowercase for easier detection
  for (const name of planetNames) {
    if (s.includes(name)) return name; // return the first planet found
  }
  return null; // return null if no planet is mentioned
}

```

- puts in the transcript

- converts to lowercase
- scans to see if planet is in the array of planets available to travel to
- send back the answer
- the dc.on message function then sees if the mentioned planet is not currently the planet that the user is on
- then changes planet

PlanetNames array:

```
const planetNames = [
  "earth", "mercury", "venus", "mars", "jupiter",
  "saturn", "uranus", "neptune", "pluto", "sun", "hole",
];
```

StartPlanetTransition:

```
// handles fade-out..planet swap... fade-in transition when the AI mentions a planet
function startPlanetTransition(newPlanet) {
  setIsTransitioning(true); // disable orbit controls and reset camera during transition

  // fade out the current planet and starfield
  let fadeOut = setInterval(() => {
    setPlanetOpacity((o) => Math.max(0, o - 0.05));
    setStarsOpacity((o) => Math.max(0, o - 0.05));
  }, 50);

  // after fade out completes, swap the planet then fade back in
  setTimeout(() => {
    clearInterval(fadeOut);
    setCurrentPlanet(newPlanet); // swap to the newly mentioned planet
  }, 500);
}
```

Current planet code:

```
const currentPlanetRef = useRef(currentPlanet); // ref to track current planet without rerendering the app when it changes
currentPlanetRef.current = currentPlanet;
```

Stores it in ref to track without re-render

```
// check if a planet was mentioned and trigger a transition
const mentionedPlanet = detectPlanetMention(answer);
if (
  mentionedPlanet &&
  mentionedPlanet !== currentPlanetRef.current
) {
  onPlanetDetected(mentionedPlanet);
}
```


-check if not current planet

Local Storage:

```
// chat history loaded from local storage
const [qaHistory, setQaHistory] = useState(() => {
  try {
    return JSON.parse(localStorage.getItem("qaHistory") || "[]");
  } catch {
    return []; //error handling in case of invalid , empty history data in local storage
  }
});
```

-load chat history

```
// adds the ai response to history and keeps it in local storage
function addQA(question, answer) {
  const item = {
    id: crypto.randomUUID(), // unique id for each history item
    ts: Date.now(), // timestamp for when it was added
    question: (question || "").trim(), //functionality to add questions too
    answer: (answer || "").trim(), //the ai reponse
  };

  setQaHistory((prev) => {
    const next = [item, ...prev];
    localStorage.setItem("qaHistory", JSON.stringify(next)); // add to local storage
    return next;
  });
}

// clears the entire chat history from state and local storage
function clearHistory() {
  localStorage.removeItem("qaHistory");
  setQaHistory([]);
}

// removes a single history item by its unique id
function removeHistoryItem(id) {
  const next = qaHistory.filter((x) => x.id !== id); // filter out the matching id
  setQaHistory(next);
  localStorage.setItem("qaHistory", JSON.stringify(next)); // update local storage
}
```

- adding ai response

-clearing full history

-clearing one item

Appendix I:

Testing Document: This document highlights the different types of testing that was conducted within the project. It also includes the test cases and whether they passed.

Testing

Black-box testing

Functional Test Case Specification

ID	Process	Description	Input	Expected result	Actual Result	Pass /Fail
1	AI voice response	Asking a valid question to system	"What planet is the largest in the solar system"	AI responds with relevant answer "Jupiter" and Transported to the planet Jupiter	Correct answer returned, "Jupiter" and transported to the planet Jupiter	Pass
2	Audio Output	User asks valid question; Audio response is generated.	Any valid input	AI tutors Audio plays clearly	The Audio played clearly	Pass
3	3D scene load	Scene loads when app opens	Open app/ load app	Scene renders correctly with default Earth state.	Scene rendered correctly with Earth	Pass
4	Chat history load	Chat history loads when window is opened	Ask valid question and then open window/ load history.	The chat history loads correctly with the conversation about the question.	Chat history loads with the correct conversation	Pass
5	Clear chat history	Chat history clears when clear button is pressed	Ask valid question and then press clear button	Clears all chat history	It cleared all chat history	Pass
6	Warp animation loads	When transporting to new planet make sure the warp animation	Ask a valid input that will transport me to a new planet e.g. "take	Animation loads, fades in and then fades out into the new planet	Animation faded out and then faded back into new planet	Pass

		loads correctly.	me to mars”			
7	New planet loads	When transporting to new planet after animation make sure it loads	Ask a valid input that will transport me to a new planet e.g. “take me to mars”	After the animation fades out the new planet is already loaded in.	The planet loads in correctly after the animation	Pass
8	Connect to AI	If connect button is pressed see if it connects to the AI	Press connect button then ask valid input	Connect button turns into disconnect button and user can communicate with AI	The connect button turned into a disconnect button and can communicate with the AI	Pass
9	Disconnect from AI	If disconnect button is pressed see if it disconnects from the AI	Press disconnect button then ask valid input	No response	No response	Pass
10	Individual chat history deletes	If delete button is pressed on an individual chat, see if it deletes	Ask valid input, then press delete button on that input in chat history	Deletes only that chat	Deletes only that chat	Pass
11	Rotate planet	User uses mouse to rotate around planet	Users Cursor drags	Rotates around planet	Rotates around planet	Pass

12	Planet orbits					
----	---------------	--	--	--	--	--

Edge Case Test Specification

ID	Edge Case	Input	Expected Result	Actual Result	Pass/Fail
1	Empty input (no question asked)	""	No response	No response	Pass
2	Extended question input	More than 30 seconds spoken input	System responds correctly	System responds correctly	Pass
3	Invalid Spoken input	Unclear speech	AI responds with error	AI responded with "Sorry I can only speak English; how else can I help you with Space related questions"	Pass
4	Irrelevant user input	"What should I eat tonight"	AI responds with redirection to correct topic.	AI redirects answer to correct topic	Pass
5	No internet	No connection/ offline	Application works as normal just without user input	Application worked as normal waiting for reconnection	Pass
6	Invalid transportation query	"Travel to the planet graphics card"	AI responds with error.	Redirection to input of real planets	Pass

7	Multiple valid transportation inputs	"Travel to mars, Jupiter, Saturn"	Travels to first planet mentioned	Travels to first planet mentioned	Pass
8	Low-volume speech input	Input is whispered very quietly (test may depend on users' microphone)	System either doesn't respond, responds with a redirect or error, responds with correct output	Lowest Volume: No response Medium unclear speech: error message Louder volume: responds with correct input	Pass
9	Background noise	Input with heavy background noise	System recognises speech or redirects user	The system recognised inputted speech but sometimes paused mid-sentence.	Fail
10	Application open in two tabs at once	Two tabs with application open at same time	System maintains individual conversations on both tabs (double speech)	Double speech	Pass

Integration testing

ID	Process	Steps	Expected Outcome	Actual Outcome	Pass/Fail
1	Full System integration	1. Connect to AI 2. Input 3. AI processing	Successful connection WebRTC with OpenAI API -> audio responses	System connected successfully, audio responses came	Pass

		4.AI response 5.UI updates	are working correctly, subtitles working correctly.	through and subtitles showed correctly.	
2	AI to audio output	1.Ask question 2.Wait for Response	Audio is played correctly without interruption	Audio response played correctly, minimal delay, clear	Pass
3	Error handling	1.disconnect internet 2.try and interact with AI			
4	AI response to chat history	1.ask AI question 2. wait for response 3.ask another question relating to first	AI responds with correct background knowledge about the conversation	AI responded with correct previous knowledge	Pass
5	Backend and API integration test	1.start express backend server 2.connect to ai	Temporary key is received securely from the backend, authentication succeeds and can chat with the AI tutor	Session key was retrieved successfully, connection was established and chat was enabled to continue	Pass
6	Transcript and subtitle integration	1.speak to AI 2.see if subtitles show	Subtitles show clearly every time when AI tutor speaks	Subtitles clearly showed every time	Pass
7	Planet detection	1.ask to travel to Jupiter	Travels to Jupiter with no errors	Travelled to Jupiter with no errors	Pass

		2.wait for response		and asked additional questions	
8	Microphone and AI	1.enable microphone in browser 2.ask question	Works correctly and AI responds or no response if no mic is set up	Worked correctly and responded	Pass
9	Delete chat history	1.delete chat history item	Chat item deletes fully	Item deleted	Pass
10	Local Storage	1.ask ai question 2.refresh application to see if still there	Chat history stays the same	Chat history stayed the same	pass

Usability Testing

<u>Participant</u>	<u>Task</u>	<u>Observation</u>	<u>Issue Identified</u>	<u>Resolution</u>
Pilot User	Complete Investigation	Participant completed the investigation successfully with minimal issues	No issues identified	No changes to the investigation process required
Pilot User	Interact with the AI tutor	Audio occasionally overlapped	Reduce response clarity	Audio handling was refined and users were instructed to only keep one tab open at a time

Feedback results from Pilot participant:

|

1. How satisfied are you with your overall experience?



2. How easy was the application to use?



3. How well did the AI understand your speech?



4. How helpful were the AI's responses?



5. What features do you feel are missing or could be improved?

The text went pretty quickly when question was answered. Also add section for past question notes like chatGPT.

6. Did you experience any technical issues?

☒ Yes

☐ No

7. If yes, please describe the issues...

Double speech, need to be careful asking questions.

8. How does this app compare to other voice or tutoring apps you've used? (1 very bad, 5 very good)



9. What did you like most about the AI Voice Tutor Space?

The information was delivered very quickly and accurately

10. What did you like least?

The text going away quickly

11. How well did you think you learnt with AI?

No answer provided.

12. How helpful was the textbooks answers?

No answer provided.

13. How easy was it to use the textbook to find answers?

No answer provided.

14. Why did you think the textbooks answers were that rating?

No answer provided.

15. How well did you think you learnt with textbooks?

No answer provided.

16. Any additional comments?

No answer provided.

Appendix J:

Raw Results/Data: Dataset derived from the investigation phase. This is shown as how many marks each participant got with a % of correct answers next to the marks.

Baseline and post-study datasets:

Baseline Tests						
Participant	Jupiter		Black Holes		Total Score	
	Score out of 10	Percentage (%)	Score out of 9	Percentage (%)	Score out of 19	Percentage (%)
1	2	20	2	22.22	4	21.05
2	1	10	1	11.11	2	10.53
3	1	10	4	44.44	5	26.32
4	4	40	2	22.22	6	31.58
5	6	60	4	44.44	10	52.63
6	2	20	6	66.67	8	42.11
7	1	10	4	44.44	5	26.32
8	1	10	2	22.22	3	15.79

Post-study tests						
Participant	Jupiter			Black Holes		
	Condition	Score out of 10	Percentage (%)	Condition	Score out of 10	Percentage (%)
1	AI	10	100	Textbook	8	80
2	Textbook	2	20	AI	7	70
3	AI	9	90	Textbook	7	70
4	Textbook	7	70	AI	8	80
5	AI	10	100	Textbook	7	70
6	Textbook	2	20	AI	9	90
7	Textbook	4	40	AI	9	90
8	AI	10	100	Textbook	5	50

Qualitative Feedback dataset:

Question	Participant					
	1	2	3	4	5	6
How satisfied are you with your overall experience?	5	5	5	5	5	5
How easy was the application to use?	5	5	5	5	5	5
How well did the AI understand your speech?	5	5	5	5	5	5
How helpful were the AI's responses?	5	5	5	5	5	5
What features do you feel are missing or could be improved?	N/A	N/A	Nothing Much	N/A	The text goes away too quickly	N/A
Did you experience any technical issues?	No	No	No	No	No	No
If yes, please describe the issues...	N/A	N/A	N/A	N/A	N/A	N/A
How does this app compare to other voice or tutoring apps you've used?	5	5	5	5	5	5
What did you like most about the AI Voice Tutor Space?	Very accurate responses and fast replies	easy to use	The AI voice sounded very human and natural	The ease of use	It gave clear, descriptive, dense answers. Sticks to the point	it used voice and had a text log
What did you like least?	N/A	N/A	Nothing Much	N/A	Nothing in particular. Maybe the text disappearing too quickly	Nothing
How well did you think you learnt with AI?	5	5	2	1	5	5
How helpful was the textbooks answers?	4	3	4	2	2	3
How easy was it to use the textbook to find answers?	2	3	3	1	1	2
Why did you think the textbooks answers were that rating?	N/A	harder to navigate	It was harder to search for answers	The textbook was difficult to find the exact information I had to.	The wording wasn't clear in the textbook. Searching through it was difficult, and the answers weren't easy to gather.	too much unnecessary information. even with keyword finder it was still difficult to find answers
How well did you think you learnt with textbooks?	2	2	5	4	2	2
Any additional comments?	N/A	N/A	From past experience, I am sure that after leaving this study, the words I have read in the textbook will stick around in my mind for sometime, I already forgot what the AI said.	I think I was able to recall the information more with the textbook as the AI did not require me to think about the answers. I just copied what was said to me and put it out of my head straight away. I was able to do an initial run of all answers much more quickly with the AI but had to revise it multiple times. With the textbook I did not require as much revision and I retained more information within the 10 minutes.	N/A	Textbooks are outdated. AI answers every question whereas textbooks may not have. AI is essentially more accessible

